



Biotechnika Info Labs

Department of Bioinformatics

Project Report On

**UTILIZING BIOINFORMATICS AND COMPUTER-AIDED
DRUG DESIGN (CADD) PIPELINES TO IDENTIFY
NOVEL THERAPEUTIC TARGETS OF MPOX**

By

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ABSTRACT

Background:

Mpox virus (MPXV), an emerging zoonotic Orthopoxvirus responsible for recurrent global outbreaks, continues to pose a significant public health challenge due to limited antiviral therapeutics and the ongoing evolution of viral lineages. Conserved viral proteins involved in essential biological processes represent promising targets for broad-spectrum antiviral development. The present study aimed to identify conserved, non-host homologous, and druggable therapeutic targets across multiple MPXV clades and to screen potential inhibitory compounds using an integrated bioinformatics and computer-aided drug discovery (CADD) workflow.

Methodology:

Twenty-eight complete MPXV genomes representing Clade I, IIa, and IIb lineages were retrieved from NCBI and subjected to comparative genomic analysis. Conserved single-copy orthogroups were identified using OrthoFinder, followed by multiple sequence alignment using MAFFT within the Galaxy platform. Functional annotation through InterProScan and UniProt, along with subtractive proteomics analysis, enabled prioritization of essential viral proteins lacking significant similarity to human proteins. Structural modelling of unresolved proteins was performed using SWISS-MODEL and validated using PROCHECK, VERIFY3D, and ERRAT. Binding pocket identification and druggability assessment were conducted using ProteinPlus. Ligand candidates selected from ZINC12 and PubChem databases were screened based on drug-likeness properties, followed by molecular docking using AutoDock Vina. Pharmacokinetic and toxicity profiling were evaluated using SwissADME and pkCSM, while Density Functional Theory (DFT)-based Frontier Molecular Orbital analysis using Python script was performed to assess electronic stability and chemical reactivity of shortlisted compounds.

Results:

Comparative genomic analysis identified 129 conserved single-copy orthogroups across genetically diverse MPXV strains. Progressive filtering through functional annotation, subtractive proteomics, structural validation, and druggability assessment reduced the dataset to 13 high-confidence target proteins containing 35 biologically relevant binding pockets. Molecular docking analysis identified multiple ligands exhibiting strong binding affinities within catalytically important regions of Poly(A) polymerase catalytic subunit, Core protein

OPG073, Virion morphogenesis protein OPG132, and Early transcription factor 82 kDa subunit. Subsequent ADMET filtering, redocking validation, and DFT-based electronic characterization prioritized five lead compounds (PubChem IDs: 3131912, 2368032, 46360552, 3410901, and 21005142) demonstrating favourable binding stability, pharmacokinetic behaviour, and electronic properties.

Conclusion:

The present study established a comprehensive computational framework for the identification of conserved antiviral targets and potential inhibitory compounds against Mpox virus. The integrated multi-stage filtering strategy effectively refined a large set of conserved proteins into a small number of biologically relevant and structurally reliable therapeutic candidates. The prioritized ligand–protein complexes demonstrated promising interaction stability and drug-like characteristics, supporting their potential for further experimental validation and future antiviral drug development against MPXV.

Keywords: *Mpox, Orthopoxvirus, Computer-aided drug design (CADD), Comparative genomics, Subtractive proteomics, Molecular docking, ADMET profiling, Density Functional Theory (DFT)*

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CHAPTER-1

INTRODUCTION

Mpox, previously known as monkeypox, is a zoonotic viral disease caused by the Mpox virus (MPXV), a member of the genus *Orthopoxvirus* within the family *Poxviridae* [1]. The virus is an enveloped, double-stranded DNA virus with a genome of approximately 197 kilobases that encodes numerous proteins involved in viral replication, transcription, host immune evasion, and virion assembly [2]. Similar to other orthopoxviruses, Mpox virus replicates in the

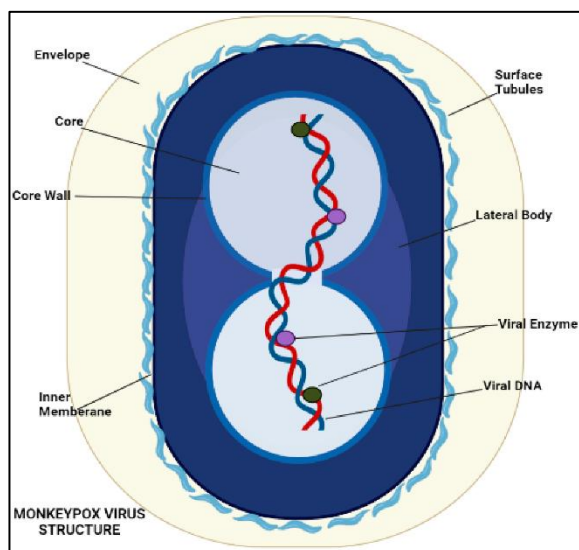


Figure 1: Mpox virus structure [67].

cytoplasm of host cells and relies on its own transcriptional and replication machinery, making several viral proteins essential for successful infection and propagation [3].

The epidemiological landscape of Mpox has evolved considerably since the first human case was reported in the Democratic Republic of Congo in 1970 [4]. Two major genetic clades of the virus have been identified: Clade I (Central African clade) and Clade II (West African clade), with multiple sub-lineages

arising through viral evolution [4]. While earlier outbreaks were largely restricted to endemic regions of Africa, the global outbreak reported in 2022 resulted in widespread transmission across more than one hundred countries [5]. Although vaccines originally developed for smallpox can provide cross-protection against Mpox infection, effective antiviral treatments remain limited, highlighting the need for new therapeutic strategies [6]

One promising strategy for antiviral discovery involves identifying conserved viral proteins that play essential roles in viral replication and survival. Proteins that remain conserved across multiple viral strains are particularly attractive drug targets because inhibitors designed against such targets may remain effective despite genetic variation among circulating viruses [7]. However, candidate targets should also exhibit minimal similarity to human proteins in order to reduce potential host toxicity and off-target effects. Computational approaches such as subtractive proteomics and comparative genomics have therefore become valuable strategies for identifying pathogen-specific therapeutic targets [8].

Advances in bioinformatics and computational biology have significantly accelerated early-stage drug discovery. Comparative genomics and orthogroup analysis enable the

identification of conserved genes across multiple viral genomes, while structural bioinformatics approaches such as homology modelling allow three-dimensional structures to be predicted when experimental structures are unavailable [9]. Structure-based drug design methods, including molecular docking and virtual screening, are widely used to predict interactions between candidate compounds and protein targets [10]. Furthermore, *in-silico* ADMET prediction tools provide early assessment of pharmacokinetic properties and toxicity risks of potential drug [11]

In the present study, an integrated bioinformatics and computer-aided drug design pipeline was employed to identify potential therapeutic targets within the Mpox virus proteome. Twenty-eight Mpox genomes representing four major clades (Ia, Ib, IIa and IIb) were analyzed to identify conserved protein orthogroups across genetically diverse viral strains. Functional annotation and subtractive proteomics were used to prioritize viral proteins that are essential for viral survival and non-homologous to human proteins. Structural modelling, binding pocket identification, and druggability assessment were then performed to evaluate the suitability of these proteins as drug targets. Finally, molecular docking and *in-silico* ADMET analysis were conducted to screen and prioritize candidate inhibitory compounds. Through this systematic computational workflow, the study aims to identify conserved Mpox viral proteins and potential small-molecule inhibitors that may serve as starting points for future antiviral development.

CHAPTER-2

LITERATURE REVIEW

The Mpox virus (MPXV), a member of the genus *Orthopoxvirus* within the *Poxviridae* family, has attracted increasing scientific attention due to its ability to cause zoonotic infections and its potential for global transmission. Historically, Mpox was considered a relatively rare disease confined to regions of Central and West Africa [12]. However, recent multi-country outbreaks have highlighted the need for improved understanding of viral biology, transmission dynamics, and potential therapeutic targets. Studies of orthopoxvirus biology have demonstrated that the Mpox genome encodes numerous proteins responsible for viral replication, transcription, host immune modulation, and virion assembly. Many of these proteins are highly conserved among orthopoxviruses, suggesting that they may represent promising targets for antiviral intervention [1]

Recent epidemiological studies have emphasized the evolving nature of Mpox virus circulation and genetic diversity. Comparative genomic analyses have identified two primary clades of the virus, commonly referred to as Clade I and Clade II, with additional sub-lineages arising due to ongoing viral evolution. The 2022 global outbreak demonstrated the capacity of the virus to spread beyond endemic regions, raising concerns about preparedness for emerging viral pathogens. As a result, several studies have focused on genomic surveillance and sequence analysis to better understand the genetic diversity of Mpox virus strains and their evolutionary relationships [4]. Comparative genomics approaches have proven valuable in identifying conserved viral genes that remain stable across multiple strains, thereby providing potential targets for therapeutic development.

In addition to genomic investigations, researchers have explored strategies for identifying pathogen-specific drug targets through subtractive proteomics. This computational approach involves comparing pathogen proteins with the host proteome in order to exclude proteins that share significant homology with human proteins [13]. By focusing on proteins that are unique to the pathogen, subtractive proteomics reduces the risk of host toxicity and increases the likelihood of identifying selective therapeutic targets. Such approaches have been successfully applied in several viral and bacterial drug discovery studies, where comparative genomics and sequence homology filtering were used to prioritize candidate proteins for further investigation [8].

Advances in structural bioinformatics have further enhanced the ability to characterize viral proteins and evaluate their suitability as drug targets. When experimentally determined

structures are unavailable, homology modelling techniques can be used to predict three-dimensional protein structures based on known templates [14]. Tools such as SWISS-MODEL and related modelling platforms have become widely used in structural biology research for generating reliable structural predictions of viral proteins [9]. Structural modelling enables researchers to identify functional domains, analyse protein folding patterns, and evaluate potential ligand-binding sites. The identification of druggable pockets within viral proteins is a critical step in structure-based drug discovery, as these pockets represent regions where small-molecule inhibitors may interact with the protein to disrupt its biological function.

Structure-based drug design and virtual screening techniques have become increasingly important tools in antiviral drug discovery. Molecular docking algorithms enable the prediction of binding interactions between proteins and small molecules by evaluating the stability and energetics of potential ligand–receptor complexes. Programs such as AutoDock Vina are widely used to perform docking simulations and estimate binding affinities between candidate compounds and protein targets [10]. Through virtual screening, large libraries of chemical compounds can be rapidly evaluated *in-silico*, significantly reducing the time and cost associated with experimental screening. These computational strategies have been applied in numerous studies aimed at identifying inhibitors of viral enzymes, structural proteins, and other essential components of viral replication machinery.

In addition to docking analysis, the evaluation of pharmacokinetic and toxicity properties is an essential step in the early stages of drug discovery. *In-silico* ADMET prediction tools allow researchers to assess properties such as absorption, distribution, metabolism, excretion, and potential toxicity of candidate compounds before experimental testing. Platforms such as SwissADME and pkCSM are commonly used to evaluate drug-likeness, bioavailability, and safety profiles of potential therapeutic molecules [15], [16]. Integrating ADMET screening into computational pipelines helps prioritize compounds that possess favorable pharmacological characteristics and reduces the likelihood of failure in later stages of drug development.

Collectively, these computational approaches have transformed modern antiviral research by enabling systematic identification of pathogen-specific targets and potential inhibitory compounds. Integrative workflows that combine comparative genomics, subtractive proteomics, structural modelling, molecular docking, and pharmacokinetic screening provide a comprehensive framework for early-stage drug discovery. Despite these advances, relatively few studies have applied such integrated pipelines specifically to Mpox virus proteins across multiple genomic clades. Therefore, the application of a comprehensive bioinformatics and

computer-aided drug design workflow offers an opportunity to identify conserved viral targets and candidate inhibitory compounds that may contribute to future therapeutic strategies against Mpox infection.

CHAPTER-3

MATERIALS AND METHODOLOGY

All computational analyses in the present investigation were conducted within a comprehensive *in-silico* framework utilizing advanced bioinformatics tools, databases, and molecular modeling platforms. The study was performed on a system powered by an Intel® Core™ Ultra 5 225U processor (12 cores, 14 threads) with 16 GB RAM, ensuring efficient handling of computational workloads. A stable broadband connection enabled uninterrupted access to online bioinformatics resources. The overall workflow encompassed sequence retrieval, proteomic characterization, functional annotation, evolutionary analysis, molecular docking, and pharmacokinetic (ADMET) evaluation of potential inhibitors.

3.1 Genome Dataset Collection

Complete genome sequences of the Mpox virus were retrieved from the National Center for Biotechnology Information GenBank database (<https://www.ncbi.nlm.nih.gov/genbank/>), a comprehensive public repository that archives annotated nucleotide sequences and associated metadata, enabling large-scale genomic analyses and comparative studies. The database provides standardized, curated access to sequence records and serves as a primary resource for molecular biology research [17]. The corresponding protein-coding sequences (CDS) for each genome were extracted in FASTA format to facilitate downstream comparative genomic analyses. A total of twenty-eight Mpox virus genomes were selected, representing four major viral clades (Ia, Ib, IIa, and IIb). To ensure balanced representation across distinct evolutionary lineages, seven genomes were included from each clade.

During dataset selection, efforts were made to incorporate both geographical and temporal diversity. Genomes originating from different countries and collection periods were included to better represent the genetic variability of circulating Mpox virus strains. Furthermore, multiple lineages within clade IIb, along with representative canonical strains, were selected to minimize bias toward any single outbreak lineage. This approach ensured that the dataset captured a broad spectrum of viral diversity.

Since clade information was not consistently annotated in GenBank records, clade assignment was performed using the Nextclade platform (<https://clades.nextstrain.org/>). Nextclade is a web-based tool developed within the Nextstrain framework that enables rapid phylogenetic classification and quality assessment of viral genomes by comparison against curated reference datasets [18]. The selected genome sequences, in FASTA format, were analysed using the Nextclade interface, which assigns each sequence to its respective clade and lineage based on sequence similarity and predefined phylogenetic criteria. For genomes classified under clade IIb, lineage-level annotations were further examined to capture finer-scale evolutionary variation within this group. Metadata associated with each genome,

including isolate name, accession number, country of origin, and year of collection, were obtained from the corresponding GenBank records and organized for subsequent analysis.

3.2 Comparative Genomics and Orthogroup Analysis

Comparative genomic analysis was performed to identify conserved protein families across the selected Mpox virus genomes. Protein-coding sequences from the twenty-eight genomes were compiled into a combined proteome dataset and used for orthology inference. Orthologous gene clustering was carried out using OrthoFinder (version 2.5.5) through the Galaxy Europe platform (<https://usegalaxy.eu/>), which enables reproducible and efficient genomic analyses.

OrthoFinder identifies orthogroups by combining sequence similarity searches with phylogenetic analysis, allowing reliable detection of evolutionarily related protein families. In this study, protein sequences in FASTA format were analyzed using the DIAMOND algorithm for fast all-versus-all similarity searches. Gene trees were inferred using the FastTree method, and multiple sequence alignments were generated with MAFFT using default settings [19]

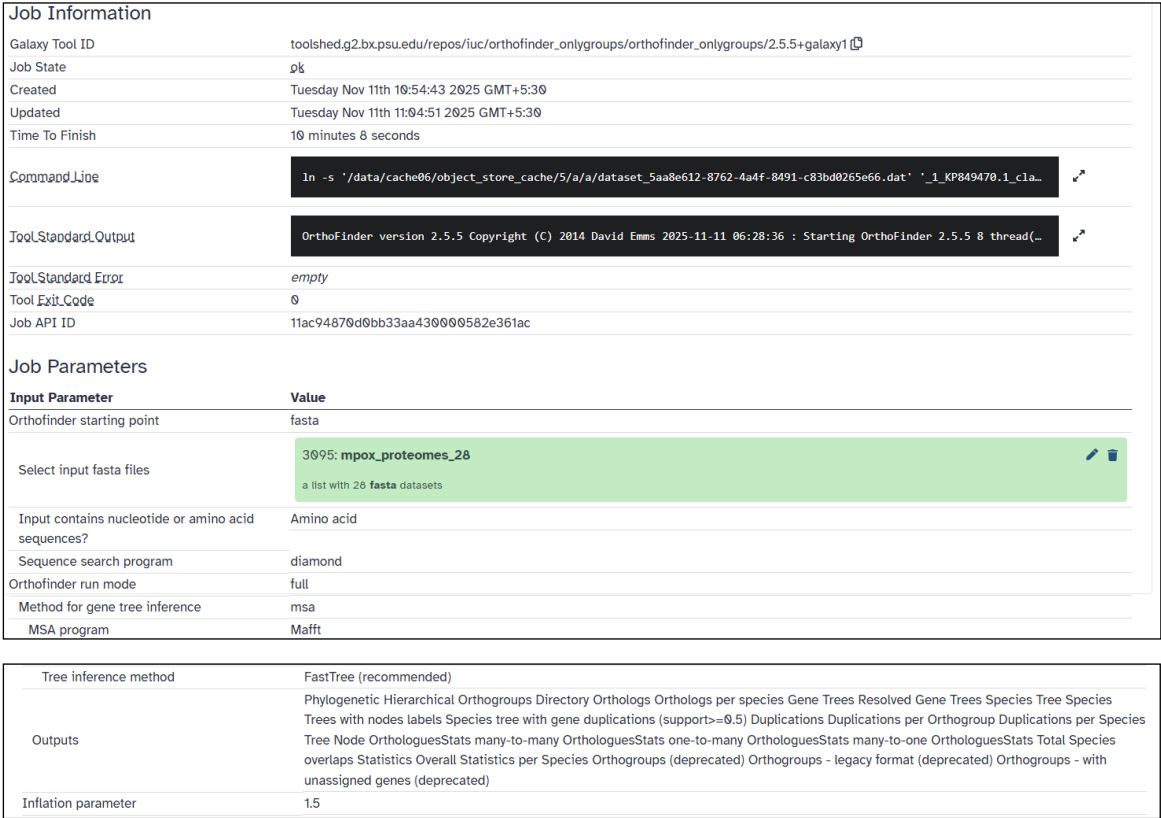


Figure 2: Orthogroup analysis performed using OrthoFinder on the Galaxy Europe platform. Job configuration and parameters used for the analysis. Output summary including orthogroup assignments and phylogenetic inference.

3.3 Functional Annotation

Following identification of conserved orthogroups through comparative genomic analysis, functional annotation was conducted to elucidate the predicted biological roles of the encoded viral proteins. Protein sequences in FASTA format derived from the orthogroup dataset were used as input for downstream functional characterization.

Functional annotation was carried out using InterProScan (<https://www.ebi.ac.uk/interpro/>), a widely used tool that integrates multiple protein signature databases to classify proteins into families and predict conserved domains and functional sites. It combines resources such as Pfam, SMART, and PROSITE, allowing reliable identification of structural features and biological functions. Through this approach, information on molecular function, domain architecture, and enzymatic activity was obtained for each protein [20].

The output generated by InterProScan was obtained in tab-separated values (TSV) format, containing protein identifiers, domain assignments, and associated functional descriptions. Since individual proteins often produced multiple domain hits, the TSV file was processed using the Datamash tool within the Galaxy Europe platform. This step grouped annotation entries and generated a non-redundant list of protein identifiers, enabling streamlined downstream analysis.

3.4 Essential Protein Identification

Following functional annotation, the conserved protein set was refined based on their relevance to the viral life cycle. A literature-based approach was used to identify proteins that are important for Mpox virus survival and replication. Annotation results were supported with information from curated databases such as UniProt (<https://www.uniprot.org/>) and published studies on orthopoxvirus biology.

UniProt provides manually reviewed and well-annotated protein information, including functional descriptions, domain details, and involvement in biological processes. This helped in confirming the roles of selected proteins and improving annotation reliability [21].

Proteins were classified as essential if they were reported to participate in key viral processes such as DNA replication, transcription, virion assembly, and host–virus interactions. Proteins without clear functional evidence or those not linked to these core processes were excluded. This step reduced the dataset to proteins with well-supported biological roles.

This filtering approach ensured that only functionally relevant proteins were carried forward, strengthening the accuracy of downstream target identification and drug discovery.

Evidence was primarily based on studies of orthopoxviruses and other members of the Poxviridae family, which share similar biological features with the Mpox virus.

3.5 Subtractive Proteomics

Subtractive proteomics analysis was carried out to identify viral proteins that do not share similarity with the human proteome. Protein sequences from the annotated Mpox dataset were compared against the human reference proteome using the BLASTp tool available through the NCBI platform (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). NCBI BLAST is a widely used sequence comparison tool that identifies regions of similarity between protein or nucleotide sequences. BLASTp, in particular, compares protein sequences and helps detect homologous relationships based on sequence alignment scores and statistical significance [22].

An E-value cutoff of $1e-5$ was applied to define significant similarity. Viral proteins that showed similarity to human proteins below this threshold were considered homologous and removed from further analysis. Only proteins without detectable similarity to the human proteome were retained as pathogen-specific candidates.

This filtering step ensures that the selected proteins are unique to the virus, thereby reducing the risk of off-target effects in the host. The final set of non-homologous proteins was then used for downstream analyses, including structural modelling, binding pocket identification, and molecular docking studies.

3.6 Structural Modelling

For proteins identified as non-homologous to the human proteome, three-dimensional (3D) structures were obtained to support structure-based analysis. Structural information is necessary for identifying ligand-binding sites and performing molecular docking studies.

The availability of experimentally resolved structures was first checked using the Protein Data Bank (PDB) (<https://www.rcsb.org/>). PDB is a global repository that stores experimentally determined protein structures obtained through techniques such as X-ray crystallography and cryo-electron microscopy. For proteins with available structures, the corresponding PDB files were retrieved and used directly [23].

For proteins without resolved structures, homology modelling was carried out using the SWISS-MODEL web server (<https://swissmodel.expasy.org/>). SWISS-MODEL is an automated platform that builds protein structures based on sequence similarity to known templates. It identifies suitable templates and generates 3D models using comparative modelling approaches [9].

To maintain model quality, sequence identity between the target protein and the selected template was used as a key criterion. Proteins with template sequence identity below 30% were excluded, as low similarity can reduce model accuracy. Only proteins meeting this threshold were retained. The final protein structures were obtained in PDB format and used for downstream analyses, including binding pocket identification and molecular docking. These models served as the basis for evaluating protein–ligand interactions in later stages of the study.

3.7 Domain Mapping

Proteins with suitable sequence identity and confirmed to be non-homologous to the human proteome were further analyzed for domain mapping to identify functionally important regions within the selected Mpx viral proteins. Conserved domains provide key information about protein function and help locate regions involved in viral replication, enzymatic activity, and structural organization.

Functional domain information was obtained using the UniProt knowledgebase and the InterPro database. UniProt was used to retrieve curated annotations, including functional descriptions, domain names, and their positions along the protein sequence. InterPro was used alongside to identify conserved domains and protein signatures by integrating multiple domain and family databases. For each protein, both the domain names and their corresponding residue positions were recorded for further analysis.

These annotations were used to describe the functional layout of each protein. The mapped residue positions helped identify catalytic regions, conserved motifs, and structurally important segments within the sequences. This information improved the understanding of protein function and highlighted regions that may act as potential ligand-binding sites. Domain mapping added an important layer of functional detail to the shortlisted Mpx proteins and guided the selection of biologically relevant regions during subsequent structural modelling and binding pocket identification.

3.8 Structural Validation and Quality Assessment

The quality and reliability of the modelled protein structures were evaluated using the SAVES (Structure Analysis and Verification Server) platform (<https://saves.mbi.ucla.edu/>). SAVES is a widely used web-based tool that combines multiple validation programs to assess different aspects of protein structure quality [24].

Several validation metrics were applied, including ERRAT, VERIFY3D, and PROCHECK. ERRAT evaluates the overall quality of the model by analysing non-bonded atomic interactions and identifying regions with possible errors [25]. VERIFY3D assesses the

compatibility between the 3D structure and its amino acid sequence by examining the residue environment [26], [27]. PROCHECK evaluates stereochemical quality by analysing backbone dihedral angles and generating Ramachandran plots to check residue conformations [28].

Together, these complementary methods provided a comprehensive assessment of structural integrity and reliability. Only models that met acceptable quality criteria were considered for further analysis. This validation step ensured that the protein structures used for binding pocket identification and molecular docking were structurally sound and suitable for structure-based studies.

3.9 Binding Pocket Identification

Binding pocket identification was carried out to locate potential ligand-binding regions within the selected Mpox viral proteins. This step is important in structure-based drug discovery, as it helps focus docking studies on biologically relevant regions of the protein.

Pocket prediction was performed using the ProteinPlus web server (<https://proteins.plus/>), with the DoGSiteScorer tool. ProteinPlus is a structure-based analysis platform that provides tools for detecting and evaluating binding sites on protein surfaces. DoGSiteScorer identifies surface cavities and assesses their suitability for ligand binding based on geometric and physicochemical features [29].

Protein structures obtained in the previous step were submitted in PDB format for analysis. DoGSiteScorer automatically detected potential pockets and assigned a druggability score to each one. Pockets with higher scores were selected, as they are more likely to bind small molecules effectively.

The selected pockets were downloaded in PDB format for use in downstream docking studies. In addition, several structural and physicochemical properties provided by ProteinPlus were recorded, including pocket volume, surface area, depth, druggability score, number of pocket atoms, hydrogen bond donors and acceptors, hydrophobic interactions, hydrophobicity ratio, apolar and polar amino acid ratios, charged residue distribution, and the presence of metal atoms. These properties helped describe the chemical environment of each binding site and were used to guide ligand selection and prioritization in subsequent molecular docking analysis.

3.10 Pocket-Domain Association Analysis

Following binding pocket identification, an additional filtering step was carried out to assess the biological relevance of the predicted pockets. The objective was to determine whether these

pockets were positioned within functionally important regions of the proteins, thereby increasing their suitability for downstream docking analysis.

Protein structures and corresponding pocket files (PDB format) obtained from the ProteinPlus platform were analysed using PyMOL V3.1, it is a widely used molecular graphics system for visualizing and analysing three-dimensional structures of proteins, nucleic acids, and small molecules. It allows detailed inspection of structural features such as binding pockets, active sites, and protein–ligand interactions. The software supports various representations (cartoon, surface, stick) and enables precise atom-level selection, distance measurement, and structural annotation. Due to its flexibility and high-quality rendering [30], PyMOL is commonly used in structural biology and computational drug discovery studies.

Domain boundaries identified in the earlier domain mapping step were used to define functional regions within each protein. Residue ranges corresponding to annotated domains were selected and visualized to enable direct comparison with predicted pocket locations. The spatial relationship between binding pockets and functional domains was evaluated by quantifying atom distribution. The total number of atoms in each predicted pocket was obtained from ProteinPlus output, while atom counts within domain regions were determined in PyMOL using selection and counting commands. This allowed estimation of how much of each pocket lies within biologically relevant regions.

A pocket–domain association metric was defined as the percentage of pocket atoms located within the annotated functional domain, calculated using the following formula:

$$\text{Pocket–domain association (\%)} = \frac{\text{Total number of pocket atoms}}{\text{Number of pocket atoms within domain}} \times 100$$

Where:

- Number of pocket atoms within domain = atoms from the predicted pocket that fall within the functional domain residue range
- Total number of pocket atoms = total atoms present in the identified pocket

Based on this metric, pockets were qualitatively classified into three categories:

- Low association: <50%
- Medium association: 50–70%
- High association: >70%

Pockets exhibiting higher association values were considered more likely to represent biologically relevant binding sites. Predicted pockets that did not overlap with annotated functional domains were excluded from further analysis.

3.11 Grid Box and Pocket Extent Calculation

For the selected pockets, spatial dimensions were calculated to define docking regions for subsequent molecular docking studies. Pocket structures were analysed in PyMOL to determine their geometric extent.

The **get_extent function** in PyMOL was used to obtain the minimum and maximum Cartesian coordinates of atoms within each pocket. Based on these coordinates, the dimensions of each pocket were calculated along the x, y, and z axes by determining the difference between maximum and minimum values.

These dimensions were used to define the docking grid box required for molecular docking simulations. Grid sizes were further adjusted and rounded to ensure consistency across all docking runs. The calculated pocket extents provided a standardized framework for defining the search space during docking and ensured accurate placement of ligands within the predicted binding regions.

3.12 Ligand Selection and Preparation

3.12.1 Compound Library Retrieval from ZINC 12

Ligands used for virtual screening were retrieved from the ZINC12 database (<https://zinc.docking.org/>), a publicly available repository of commercially accessible compounds designed for virtual screening applications. ZINC12 provides an advanced filtering interface that enables selection of compounds based on physicochemical and drug-likeness properties, making it suitable for targeted ligand library construction [31]. Compound selection was performed using multiple parameters, including molecular weight, xLogP, net charge, rotatable bonds, polar surface area, hydrogen bond donors, hydrogen bond acceptors, polar and apolar desolvation energies, and purchasability status. These filters ensured that the selected compounds fall within drug-like chemical space and are feasible for downstream experimental validation.

3.12.2 Integration with PubChem for Structural Data

To obtain structural information, compound identifiers (ZINC IDs) were mapped to corresponding entries in PubChem database (<https://pubchem.ncbi.nlm.nih.gov/>). PubChem is a publicly available chemical database maintained by the National Center for Biotechnology Information (NCBI). It provides detailed information on small molecules, including chemical structures, physicochemical properties, biological activities, and associated bioassay data. Each compound is assigned a unique Compound ID (CID), which facilitates easy retrieval and cross-referencing across datasets. PubChem is widely used in bioinformatics and drug discovery

studies for obtaining 2D/3D structures, SMILES notation, and compound-related metadata required for molecular docking and *in-silico* analysis [32].

The 3D structures of selected compounds were downloaded in Structure Data File (SDF) format, along with their SMILES representations. The SDF files were used for molecular docking and structural preparation, while SMILES strings were retained for subsequent ADMET analysis, as they provide a compact and versatile representation of molecular structure suitable for predictive modeling workflows.

3.12.3 Pocket-Guided Ligand Filtering Strategy

Ligand selection was not performed randomly but was guided by structural characteristics of the predicted binding pockets obtained from ProteinPlus analysis. Prior to compound filtering, pocket properties such as hydrogen bond donor and acceptor capacity, hydrophobic interaction density, polarity distribution, amino acid composition, pocket volume, surface area, and depth were evaluated.

These structural features were used to tailor ligand selection criteria, ensuring physicochemical complementarity between ligands and binding sites. This approach enhances the likelihood of identifying biologically relevant interactions and improves the efficiency of virtual screening. For each selected binding pocket, five candidate ligands were shortlisted, ensuring a balanced yet computationally manageable screening set.

3.12.4 Mapping Pocket Properties to Drug-Likeness Filters

Ligand filtering parameters in ZINC 12 were selected based on established drug-likeness principles and adjusted according to pocket characteristics.

Molecular weight, hydrogen bond donors (HBD), hydrogen bond acceptors (HBA), and lipophilicity (xLogP) were primarily guided by Lipinski's Rule of Five [33], which correlates these properties with oral bioavailability and membrane permeability. Ligands were restricted to $\text{HBD} \leq 5$ and $\text{HBA} \leq 10$, while molecular weight was generally maintained below 500 g/mol to ensure drug-like behaviour.

Hydrophobicity parameters were aligned with pocket composition. Hydrophobic pockets were screened using ligands with moderate lipophilicity ($\text{xLogP} \leq 5$), ensuring compatibility without compromising solubility. Rotatable bond limits (≤ 10) were applied based on Veber's criteria [34], which emphasize the importance of molecular flexibility in oral bioavailability.

Polar surface area (PSA) was restricted to $\leq 140 \text{ \AA}^2$, following findings by [35], to maintain a balance between permeability and hydrogen bonding capacity. $\text{PSA} \leq 140 \text{ \AA}^2$

generally have good oral bioavailability while $PSA \leq 90 \text{ \AA}^2$ have increased likelihood of blood-brain barrier penetration.

Desolvation parameters were incorporated based on [36], highlighting the role of solvation energetics in ligand binding. Polar pockets were matched with ligands having balanced polarity, while hydrophobic pockets were screened with compounds exhibiting favourable apolar desolvation characteristics. Net molecular charge was maintained neutral to optimize membrane permeability and electrostatic compatibility within binding pockets.

3.12.5 Final Ligand Selection for Virtual Screening

The final ligand set was curated to prioritize compounds with favorable oral bioavailability, structural compatibility with binding pockets, and drug-like physicochemical properties. The filtering strategy ensured that selected ligands occupy a chemically relevant and pharmacologically viable space.

For each retained binding pocket, five ligands were selected, resulting in a focused compound library suitable for efficient and reliable virtual screening. This systematic and structure-guided ligand selection approach strengthens the robustness of downstream docking and ADMET analyses, while maintaining biological relevance and translational feasibility.

3.13 Molecular Docking and Hit Prioritization

3.13.1 Binding Pocket Volume Filtering

To improve the reliability of molecular docking, predicted binding pockets were filtered according to cavity volume. Pockets with volumes $<300 \text{ \AA}^3$ were excluded due to limited capacity for ligand accommodation, while pockets $>1000 \text{ \AA}^3$ were removed because excessively large cavities may promote non-specific interactions and inflated docking scores. Only pockets within the defined volume range were retained for subsequent virtual screening analyses.

In addition, only binding pockets previously classified with high or medium druggability scores during the pocket–domain association analysis were considered for downstream screening. Pockets exhibiting higher pocket–domain association values were considered more likely to represent biologically relevant binding regions. Predicted pockets that did not overlap with annotated functional domains were excluded from further analysis to ensure functional relevance of the selected docking sites.

3.13.2 Protein Preparation for Docking

Protein structures were prepared using MGL Tools 1.5.7 to ensure compatibility with AutoDock Vina, following established docking protocols. MGL Tools is a widely used software suite for preparing protein and ligand structures, supporting key preprocessing steps such as hydrogen addition, charge assignment, atom type definition, and conversion into docking-compatible formats such as PDBQT [37].

During preparation, all water molecules were removed, as non-conserved solvent molecules can interfere with ligand binding. Heteroatoms, including co-crystallized ligands, ions, and other non-essential molecules, were also excluded to minimize steric bias. Polar hydrogens were added to represent hydrogen bonding interactions accurately, and Kollman charges were assigned to enable proper calculation of electrostatic interactions within the AutoDock framework. AutoDock4 atom types were then defined for all atoms. The final processed protein structures were saved in PDBQT format, which contains atomic coordinates, partial charges, and atom type information required for docking simulations.

3.13.3 Ligand Preparation and Optimization

Ligand preparation was performed using Open Babel in combination with MGL Tools to obtain structures suitable for molecular docking. Open Babel is an open-source chemical toolbox widely used for file format conversion, structure editing, hydrogen addition, and geometry optimization in computational workflows [38].

Ligand structures obtained in SDF format were first converted to MOL2 format using Open Babel. During this step, hydrogen atoms were added and geometry optimization was carried out using the Universal Force Field (UFF) to generate stable conformations. The optimized ligands were then processed using MGL Tools, where Gasteiger partial charges were assigned and AutoDock-compatible atom types were defined. Finally, the ligands were saved in PDBQT format for use in docking simulations.

3.13.4 Grid Box Definition and Docking Protocol

Binding pockets identified during structural analysis were utilized to define precise docking search spaces for molecular docking simulations. Spatial characterization of each pocket was performed in PyMOL, where the geometric extent of pocket residues was used to determine grid parameters.

For each selected pocket, the minimum and maximum Cartesian coordinates were obtained, defining the three-dimensional boundaries of the binding region:

$$\text{Min} = [X_{\text{min}}, Y_{\text{min}}, Z_{\text{min}}], \text{Max} = [X_{\text{max}}, Y_{\text{max}}, Z_{\text{max}}]$$

The geometric center of the pocket, corresponding to the grid box center, was calculated using:

$$X_c = \frac{X_{\min} + X_{\max}}{2}, Y_c = \frac{Y_{\min} + Y_{\max}}{2}, Z_c = \frac{Z_{\min} + Z_{\max}}{2}$$

Grid box dimensions along each axis were computed as:

$$S_x = X_{\max} - X_{\min}, S_y = Y_{\max} - Y_{\min}, S_z = Z_{\max} - Z_{\min}$$

To ensure complete coverage of the binding site and accommodate ligand flexibility, a buffer of approximately 4–6 Å was incorporated into each dimension. This adjustment allowed sufficient space for ligand exploration while maintaining focus within the predicted pocket region.

3.13.5 Molecular Docking Protocol

Docking simulations were performed using a standardized configuration to ensure consistency across all protein–ligand systems. The exhaustiveness parameter was set to 8, providing a balance between computational efficiency and conformational sampling depth. A maximum of 9 binding modes was generated for each docking run, enabling evaluation of multiple ligand orientations. An energy range of 3 kcal/mol was applied to retain only energetically relevant poses for subsequent analysis. This systematic and geometry-driven grid definition strategy ensured accurate localization of docking search spaces, minimized bias across simulations, and provided a consistent framework for reliable comparative analysis.

3.13.6 Post-Docking Analysis, Limitations and Reliability Considerations

Docking results were analyzed based on predicted binding affinities and binding pose localization within the target pocket. Compounds were ranked according to their binding energy values, and lower-affinity binders were excluded. A threshold of ≤ -8.0 kcal/mol was used to identify strong binding interactions, consistent with commonly accepted criteria in structure-based drug discovery studies. Ligands meeting this threshold were shortlisted for subsequent ADMET evaluation.

Majority of the protein structures used in this study were derived from homology modeling, which may introduce structural uncertainties, particularly in flexible regions and surface-exposed residues. These limitations can influence docking accuracy and binding energy predictions. During post-docking analysis, some ligands were observed to bind outside the predicted pocket or localize on the protein surface, indicating potential false-positive interactions.

3.14 *In-Silico* ADMET Profiling

Ligands that met the docking score threshold (≤ -8.0 kcal/mol) were subjected to *in-silico* ADMET evaluation to assess their pharmacokinetic suitability and potential toxicity risks. The SMILES representations of the shortlisted compounds were used as input for prediction. ADMET profiling was performed using SwissADME and pkCSM, two widely used web-based platforms.

SwissADME (<https://www.swissadme.ch/>) is a web-based tool used to evaluate physicochemical properties, pharmacokinetics, drug-likeness, and medicinal chemistry properties of small molecules. It helps in assessing oral bioavailability, lipophilicity, and potential drug interactions during early-stage drug discovery [39]. pkCSM (<https://biosig.lab.uq.edu.au/pkcsml/>) was employed to predict toxicity-related properties such as hepatotoxicity, hERG inhibition, mutagenicity, and clearance [40].

3.14.1 Physicochemical and Pharmacokinetic Parameters (SwissADME)

Key physicochemical and pharmacokinetic descriptors relevant to oral drug-likeness were evaluated. Molecular weight, lipophilicity (LogP), and hydrogen bonding capacity were assessed to ensure compliance with established drug-likeness criteria. Topological polar surface area (TPSA) was analyzed as an indicator of membrane permeability and absorption potential.

Gastrointestinal (GI) absorption was used as a primary indicator of oral bioavailability, while blood–brain barrier (BBB) permeability predictions were considered to evaluate potential central nervous system exposure. Cytochrome P450 (CYP) inhibition profiles were examined to assess the likelihood of metabolic interactions, particularly with major isoforms including CYP1A2, CYP2C19, CYP2C9, CYP2D6, and CYP3A4.

Additionally, compounds were screened for pan-assay interference compounds (PAINS) alerts to eliminate molecules with a high probability of non-specific or assay-interfering behavior. Bioavailability scores were also considered as an overall indicator of oral drug-likeness.

3.14.2 Toxicity and Clearance Assessment (pkCSM)

Toxicological and pharmacokinetic risk parameters were further evaluated using pkCSM. Total clearance was analyzed to estimate the rate of compound elimination and potential systemic exposure.

Cardiotoxicity risk was assessed using hERG inhibition models (hERG I and hERG II), which predict the likelihood of potassium channel blockade associated with cardiac arrhythmias. Mutagenicity was evaluated using the AMES toxicity model, while hepatotoxicity predictions were used to assess potential liver-related adverse effects.

3.14.3 ADMET Based Risk Categorization

Based on the combined outputs from SwissADME and pkCSM, ligands were systematically classified into four risk categories:

- **Low Risk:** Compounds exhibiting favorable physicochemical properties, high predicted GI absorption, minimal CYP inhibition, absence of hERG inhibition, negative AMES toxicity, acceptable clearance, and no predicted hepatotoxicity.
- **Moderate Risk:** Compounds retaining overall drug-like characteristics but showing limited and manageable risk factors, such as predicted hepatotoxicity or a small number of CYP interactions, without major cardiotoxicity alerts.
- **Moderate–High Risk:** Compounds presenting multiple cautionary indicators, including increased CYP inhibition, predicted mutagenicity, hepatotoxicity, or hERG II positivity.
- **High Risk:** Compounds associated with significant toxicity concerns, such as strong hERG inhibition, AMES positivity, extensive CYP interactions, or unfavorable clearance profiles.

3.15 Prioritization, Cross-Validation and Sensitivity Checks

To improve the reliability of docking results, ligands that satisfied both docking score and ADMET criteria were subjected to a secondary validation step through re-docking using AutoDock Vina with refined parameters. In the initial docking stage, larger grid boxes were used to ensure full coverage of the predicted binding pockets. However, such broad search spaces can allow ligands to bind outside the functional cavity, leading to non-specific or surface-level interactions. To address this, grid boxes were refined during re-docking by centering them precisely on the predicted pocket coordinates and reducing their size to better restrict ligand sampling within the active site.

Grid parameters were recalculated based on the minimum and maximum coordinates of the selected pocket residues. A grid size of $20 \times 20 \times 20$ Å was applied, providing sufficient coverage of the binding cavity while excluding irrelevant regions. The exhaustiveness parameter was increased to 32 to allow more thorough conformational sampling and to improve reproducibility. After re-docking, ligands with binding energies ≤ -8.0 kcal/mol were retained for further prioritization.

3.16 Frontier Molecular Orbital (FMO) Analysis Using DFT

The electronic properties of the selected top ligands were investigated through Frontier Molecular Orbital (FMO) analysis employing Density Functional Theory (DFT)-based computational approaches. FMO analysis provides valuable insights into molecular reactivity, stability, and charge transfer behaviour by evaluating the energies of frontier orbitals. DFT, a widely adopted quantum mechanical method, offers an optimal balance between computational efficiency and accuracy, making it suitable for studying the electronic structure of biologically relevant molecules.

The computational workflow was implemented using a Python-based pipeline integrating scientific libraries such as *NumPy* for numerical computations, *Pandas* for data handling, and *Matplotlib* for visualization, along with cheminformatics tools for molecular processing and descriptor calculation. Initial ligand structures were generated from SMILES notation and subjected to geometry optimization using the MMFF94 force field to obtain energetically stable conformations. The optimized structures were subsequently used for DFT calculations employing the B3LYP exchange–correlation functional with the STO-3G basis set. In instances where convergence was not achieved, calculations were repeated using the PBE functional to ensure computational robustness.

The code utilized in this study was adapted from a publicly available Kaggle repository, and the complete implementation is provided in Appendix, enabling reproducibility and systematic evaluation of quantum chemical parameters for the shortlisted ligands. Additionally, HOMO-LUMO 2d slice images were generated for visualization purposes.

3.16.1 HOMO-LUMO and Energy Gap

The Highest Occupied Molecular Orbital (HOMO) and Lowest Unoccupied Molecular Orbital (LUMO) describe a molecule's ability to donate and accept electrons, respectively. The energy difference between them, known as the HOMO–LUMO gap (ΔE), is a key indicator of molecular stability and reactivity. A smaller ΔE suggests higher reactivity and lower stability, which can enhance interactions with biological targets, while a larger ΔE indicates greater stability and reduced reactivity. These properties play a crucial role in determining ligand behaviour and binding potential [41]. The HOMO–LUMO energy gap (ΔE) was computed as [42].

$$\text{Energy Gap } (\Delta E) = E_{LUMO} - E_{HOMO} \text{ --- (1)}$$

3.16.2 Global Reactivity Descriptors: Hardness and Softness

Chemical hardness (η) and softness (S) are key reactivity descriptors derived from HOMO and LUMO energies. Hardness is defined as half of the HOMO–LUMO gap, while softness is its inverse. Molecules with higher hardness (larger energy gap) are generally more stable and less reactive, whereas softer molecules (smaller gap) are more polarizable and chemically reactive. In ligand–protein interactions, softer ligands tend to better adapt to the electronic environment of the binding site, improving interaction potential [42], [43], [44]

$$\text{Chemical hardness } (\eta) = \frac{E_{LUMO} - E_{HOMO}}{2} \text{ --- (2)}$$

$$\text{Chemical softness } (S) = \frac{1}{\eta} \text{ --- (3)}$$

3.16.3 Electronegativity and Electrophilicity

Electronegativity (χ) and electrophilicity index (ω) are important descriptors that reflect a molecule's tendency to attract and accept electrons. Electronegativity indicates electron-attracting ability, while electrophilicity measures the stabilization gained upon electron acceptance. Higher electrophilicity enhances interactions with nucleophilic residues in protein binding sites, while electronegativity influences charge distribution and binding orientation. Together, these parameters are useful for evaluating ligand reactivity, efficiency, and binding potential [42], [44].

$$\text{Electronegativity } (\chi) = -\frac{E_{HOMO} + E_{LUMO}}{2} \text{ --- (4)}$$

$$\text{Electrophilicity } (\omega) = \frac{\chi^2}{2\eta} \text{ --- (5)}$$

From the above equations 1,2,3, 4 and 5:

- E_{HOMO} : Energy of the Highest Occupied Molecular Orbital
- E_{LUMO} : Energy of the Lowest Unoccupied Molecular Orbital
- ΔE : HOMO–LUMO energy gap
- η : Chemical hardness
- S : Chemical softness
- χ : Electronegativity
- ω : Electrophilicity index

CHAPTER-4

RESULTS AND DISCUSSION

4.1 Genome Dataset Collection and Clade Distribution

A total of twenty-eight Mpox virus genomes were successfully curated using genbank for comparative genomic analysis, encompassing all four major viral clades (Ia, Ib, IIa, and IIb). The dataset was designed to maintain equal representation across clades, with seven genomes included from each group, thereby ensuring balanced evolutionary coverage.

Clade assignment performed using the Nextclade platform confirmed that all selected genomes were accurately classified into their respective clades. In addition to clade-level grouping, (figure 3) lineage analysis within clade IIb revealed the presence of multiple distinct lineages, reflecting the ongoing diversification of this clade. This observation is consistent with the increased transmission and global spread associated with clade IIb in recent outbreaks.

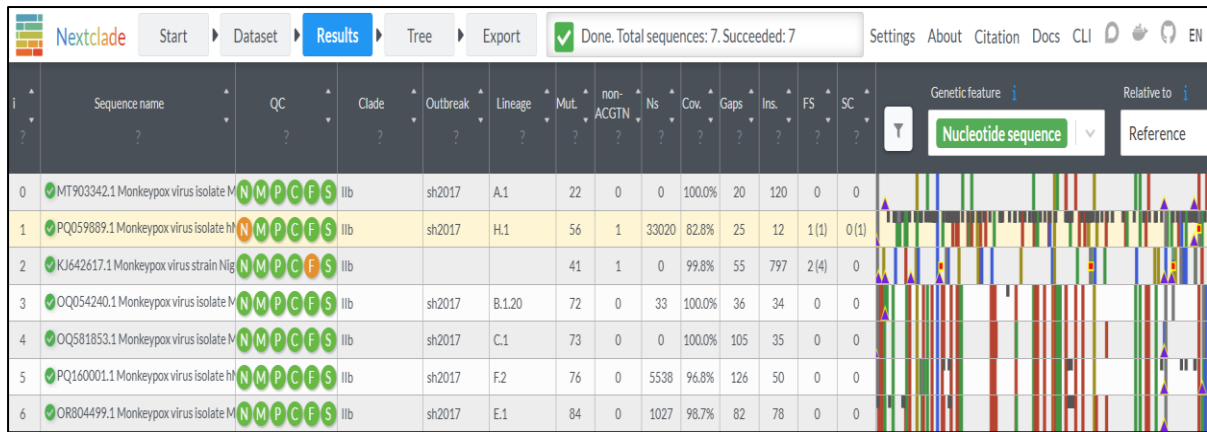


Figure 3: Clade and lineage assignment of selected Mpox virus genomes using the Nextclade platform, 7 genomes belonging to clade IIb with their lineages.

Phylogenetic analysis (figure 4) further supported the clade classification results. The generated phylogenetic tree demonstrated clear clustering of genomes according to their assigned clades, with distinct separation between clade I (Ia, Ib) and clade II (IIa, IIb) groups. Notably, genomes within clade IIb exhibited greater branching complexity compared to other clades, indicating higher genetic variability within this group.

The selected dataset captured a broad range of geographical and temporal diversity. Genomes included isolates from multiple regions, including Africa, Europe, Asia, and North America, spanning several decades. This wide representation enhances the robustness of downstream analyses by reducing bias toward specific outbreak events or geographic regions.

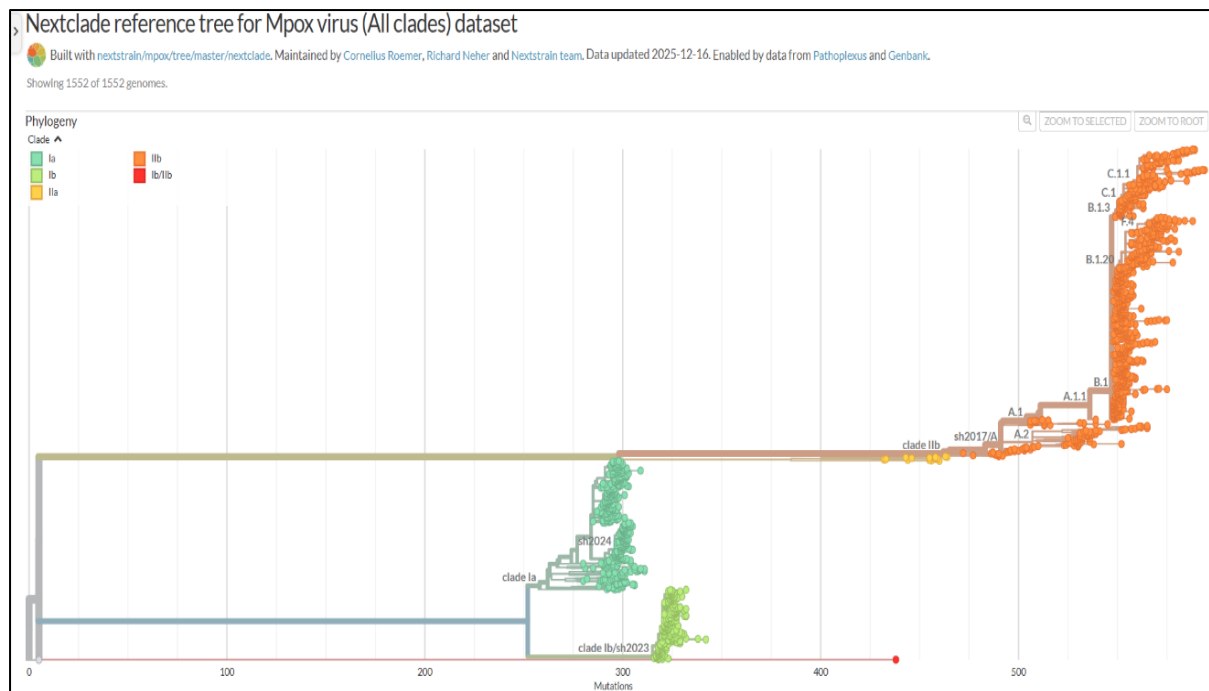


Figure 4: Phylogenetic relationships among selected Mpox virus genomes generated using the Nextstrain framework. The tree illustrates distinct clustering into four major clades and highlights increased diversity within clade IIb.

Table 1: Summary of Mpox virus genomes used in this study, including genome identifiers, accession numbers, clade classification, geographic origin, and year of isolation.

Genome ID	Virus Strain	Accession Number	Clade	Country	Year
G1	Zaire-96-I-16	NC_003310.1	Ia	Central Africa, DRC	1996
G2	Zaire_1979-005	DQ011155.1	Ia	Democratic Republic of Congo (Zaire)	1979
G3	Zaire-96-I-16	AF380138.1	Ia	Democratic Republic of Congo (Zaire)	1996
G4	Kamituga/2024/KAB-001	OQ729808	Ia	Democratic Republic of Congo	2024
G5	18MPX0083S	PQ221485	Ia	DRC, Haut-Uele, Isiro	2018
G6	18MPX0170C	PQ221490	Ia	DRC, Bas-Uele, Buta	2018
G7	22hMPX035V	PQ221499.1	Ia	DRC, Maniema, Tunda	2022
G8	MPXV/Ib/Germany/2024/ON/RKI1214	PV298252.1	Ib	Germany	2024
G9	MPXV_ITA_202505_INM I-PTV	PV844899.1	Ib	Italy	2025
G10	MPXV_USA_2025_NH01 00	PV294984.1	Ib	United States of America	2025
G11	24MPX0223C	PP601218.1	Ib	DRC, Sub-Kivu, Kamituga	2023
G12	CU-ID24110	PQ240043.1	Ib	Thailand	2024
G13	MPXV_UK_2024_0013	PQ762586.1	Ib	United Kingdom	2024
G14	MPXV/Ib/Germany/2024/RKI1142	PQ499044.2	Ib	Germany	2024

G15	Cote d'Ivoire_1971	KP849470.1	Ila	Ivory Coast, West Africa	1971
G16	Liberia_1970_184	DQ011156.1	Ila	Liberia	1984
G17	Sierra Leone	AY741551.1	Ila	Sierra	1970
G18	MPXV-WRAIR7-61	AY603973.1	Ila	Denmark	1961
G19	COP-58	AY753185.1	Ila	United States, Wisconsin	2003
G20	PCH	KJ642616.1	Ila	France, Paris	1968
G21	MPXV-USA2003_099_Gambian_Rat	MT903346.1	Ila	United States of America	2003
G22	MPV/PT0732/2023	OR804499.1	Ilb (E.1 lineage)	Portugal	2023
G23	hMpxV/USA/CA-CDPH-1M1000489/2023	PQ160001.1	Ilb (F.2 lineage)	United States of America	2023
G24	MPXV/human/Taiwan/110-093831/2023	OQ581853.1	Ilb (C.1 lineage)	Taiwan	2023
G25	MPXV_USA_2022_TX0027	OQ054240.1	Ilb (B.1.20 lineage)	United States of America	2022
G26	Nigeria-SE-1971	KJ642617.1	Ilb	Nigeria	1971
G27	hMpxV/Austria/MUW1721722/2024	PQ059889.1	Ilb (H.1 lineage)	Austria	2024
G28	MPXV-Singapore	MT903342.1	Ilb (A.1 lineage)	Singapore	2020

4.2 Orthogroup Analysis and Identification of Conserved Proteins

Orthogroup analysis figure 5 of the twenty-eight Mpox virus proteomes resulted in the clustering of protein sequences into 198 orthogroups, reflecting both conserved and variable protein families across the dataset. A total of 4,946 genes were analyzed, of which 4,931 genes (99.7%) were successfully assigned to orthogroups, indicating a high level of completeness and consistency in the dataset.

Out of the identified orthogroups, 129 orthogroups were conserved across all twenty-eight genomes, representing the core proteome shared among all analyzed Mpox virus strains. Notably, all conserved orthogroups were identified as single-copy orthogroups, suggesting that these proteins are highly conserved without duplication events and are likely to play essential roles in the viral life cycle.

Number of species	28
Number of species	28
Number of genes	4946
Number of genes in orthogroups	4931
Number of unassigned genes	15
Percentage of genes in orthogroups	99.7
Percentage of unassigned genes	0.3
Number of orthogroups	198
Number of species-specific orthogroups	0
Number of genes in species-specific orthogroups	0
Percentage of genes in species-specific orthogroups	0.0
Mean orthogroup size	24.9
Median orthogroup size	28.0
G50 (assigned genes)	28
G50 (all genes)	28

O50 (assigned genes)	86
O50 (all genes)	86
Number of orthogroups with all species present	129
Number of single-copy orthogroups	129
Date	2025-11-11
Orthogroups file	Orthogroups.tsv

Figure 5: Summary statistics of orthogroup analysis generated using OrthoFinder on the Galaxy Europe platform, showing the number of species, genes, orthogroups, and conserved single-copy orthogroups.

The high percentage of genes assigned to orthogroups, along with the absence of species-specific orthogroups, indicates strong conservation across the Mpox virus genomes despite differences in clade, geographic origin, and time of isolation. The identification of 129 conserved orthogroups further supports the presence of a stable core set of viral proteins required for fundamental biological processes. These conserved proteins are expected to be involved in essential viral functions such as genome replication, transcription, enzymatic activity, and virion assembly. Their presence across all genomes suggests that they are under strong evolutionary constraint, making them less tolerant to mutations and therefore suitable candidates for therapeutic targeting.

In contrast, the remaining orthogroups that were not universally conserved likely represent accessory or variable proteins, which may contribute to strain-specific characteristics such as host adaptation, immune evasion, or differences in virulence. To facilitate downstream analysis, only the conserved orthogroups present across all genomes were retained. Representative protein sequences from these orthogroups were extracted and used for subsequent functional annotation and target prioritization. Overall, orthogroup analysis enabled the systematic reduction of the Mpox virus proteome into a well-defined set of conserved

protein candidates, providing a strong foundation for identifying potential broad-spectrum antiviral targets.

4.3 Functional Annotation of Conserved Proteins

A total of 129 conserved protein sequences, identified from orthogroup analysis, were subjected to functional annotation using the InterProScan platform. The analysis generated detailed functional information for each protein, including predicted domains, conserved motifs, and associated biological roles. The annotation results were obtained in tab-separated format and subsequently processed to produce a non-redundant dataset, ensuring that each conserved protein was uniquely represented along with its corresponding functional description. This step minimized redundancy arising from multiple domain hits per protein and enabled streamlined downstream analysis.

Functional characterization revealed that the conserved Mpox virus proteins are involved in a wide range of essential biological processes. These include proteins associated with viral DNA replication, transcriptional regulation, enzymatic activity, and structural organization of the virion. The consistent presence of these functional categories across all conserved proteins highlights their importance in maintaining fundamental viral processes. At this stage, no filtering or exclusion of proteins was performed. The objective of functional annotation was to assign biological roles to all conserved proteins rather than to prioritize or eliminate candidates. This ensures that downstream filtering steps are based on biologically informed criteria rather than preliminary assumptions.

The complete list of annotated proteins, including protein identifiers and corresponding functional descriptions is provided in Supplementary Table S1. Overall, functional annotation established a foundational understanding of the conserved Mpox virus proteome and provided essential context for subsequent filtering steps, including host non-homology screening and target prioritization.

4.4 Essential Protein Identification

Following functional annotation, the conserved protein dataset was further refined based on biological relevance using a literature-driven approach. From the initial set of 129 conserved proteins, a total of 27 proteins were identified as essential, based on their reported involvement in critical viral processes.

The selected proteins were consistently associated with key functions such as DNA replication, transcriptional regulation, enzymatic activity, virion assembly, and host–virus interactions. Supporting evidence for these proteins was obtained from published studies on

orthopoxviruses and other members of the Poxviridae family, ensuring that the prioritization was grounded in established biological knowledge. The reduction from 129 conserved proteins to 27 essential candidates highlights the effectiveness of literature-based filtering in identifying biologically significant targets. This step enabled the exclusion of proteins with unclear or non-essential roles, thereby focusing subsequent analyses on proteins with well-defined functional importance.

Importantly, many of the identified proteins have been previously reported to play central roles in viral replication and structural organization, reinforcing their relevance as potential antiviral targets. Their conservation across multiple Mpox virus clades further supports their suitability for broad-spectrum therapeutic intervention.

The complete list of essential proteins, including protein NCBI identifiers, Uniprot identifiers functional annotations, FASTA sequences, and supporting literature references DOI, is provided in Supplementary Table S2. Overall, this step resulted in a refined and biologically validated set of protein targets, which were subsequently subjected to subtractive proteomics analysis.

4.5 Subtractive Proteomics Analysis

Subtractive proteomics analysis was carried out to identify viral proteins that do not share significant similarity with the human proteome. The set of 27 essential Mpox virus proteins was subjected to BLASTp analysis against the human reference proteome using an E-value threshold of $1e-5$.

The analysis identified five viral proteins that exhibited significant sequence similarity to human proteins and were therefore excluded to minimize the risk of host off-target effects. The excluded proteins were AAL40494.1 (C10L), AAL40515.1 (F8L), AAL40523.1 (I4L), AAL40527.1 (I8R), and AAL40593.1 (A25R).

Following this filtering step, a total of 22 proteins were retained (Table 2). These proteins represent a refined set that is conserved across Mpox virus genomes, functionally essential for viral survival, and non-homologous to the human proteome. The exclusion of proteins with human similarity is a critical step in drug target identification, as it reduces the likelihood of cross-reactivity and potential toxicity in the host. By focusing on pathogen-specific proteins, this approach enhances the selectivity and safety profile of potential therapeutic interventions.

Table 2: List of 22 proteins which are conserved, functionally essential for Mpox and non-homologous to human proteome.

Serial No.	NCBI ID	Uniprot ID	Protein Name	Functional Annotation	Reference
1	AAL40508.1	Q8V530	Poly(A) polymerase catalytic subunit	Polymerase that creates the 3'-poly(A) tail of mRNA's.	[45]
2	AAL40511.1	Q8V527	DNA-directed RNA polymerase 30 kDa polypeptide	DNA-dependent RNA polymerase carries out viral gene expression in the host cytoplasm.	[46]
3	AAL40516.1	Q8V522	Sulfhydryl oxidase	Role in morphogenesis. Vaccinia virus E10R protein is associated with the membranes of intracellular mature virions and has a role in morphogenesis.	[47]
4	AAL40517.1	Q8V521	Core protein OPG073	Virion morphogenesis	[48], [49]
5	AAL40522.1	Q8V516	Protein OPG079	Facilitates viral DNA replication by binding single-stranded DNA with high affinity and concentrating within cytoplasmic replication factories. It resolves secondary structures such as loops and hairpins, thereby minimizing polymerase stalling during genome elongation.	[50]
6	AAL40526.1	Q8V512	I7L	Viral protease	[51]
7	AAL40528.1	Q8V510	Metalloendopeptidase	Involved in maturation of some viral proteins by processing them preferentially at Ala-Gly- -Ser/Thr/Lys motifs. Does not seem to be responsible for the cleavage of major core proteins.	[52]
8	AAL40541.1	Q8V4Z9	Core protein VP8S	Major core structural protein	[53]
9	AAL40550.1	Q77HN0	H2R	Vaccinia virus entry and cell-cell fusion	[54]

10	AAL40554.1	Q8V4Y9	DNA topoisomerase I	Relieves DNA supercoiling generated during replication and transcription by transient single-strand cleavage and religation. A catalytic tyrosine mediates site-specific transesterification at 5'-[CT]CCTT, forming a 3'-phosphotyrosyl intermediate and permitting strand rotation to dissipate torsional stress, followed by restoration of the phosphodiester backbone.	[55]
11	AAL40565.1	Q8V4X8	E10R	mRNA decapping enzymes. D9 and D10 belong to the Nudix hydrolase superfamily of enzymes. Nudix enzymes catalyze the hydrolysis of nucleoside diphosphates linked to any moiety.	[56]
12	AAL40564.1	Q8V4X9	E9R	mRNA decapping enzymes. D9 and D10 belong to the Nudix hydrolase superfamily of enzymes. Nudix enzymes catalyze the hydrolysis of nucleoside diphosphates linked to any moiety.	[56]
13	AAL40570.1	Q8V4X4	Viral late gene transcription factor 3	Acts with RNA polymerase to initiate transcription from late gene promoters.	[57]
14	AAL40572.1	Q8V4X2	Major core protein OPG129	Major component of the virion core that undergoes proteolytic processing during the immature virion (IV) to mature virion (MV) transition. Essential for the formation of a structurally normal core.	[58]
15	AAL40574.1	Q8V4X0	DNA-directed RNA polymerase 19 kDa subunit	Part of the DNA-dependent RNA polymerase which catalyzes the transcription of viral DNA into RNA using the four ribonucleoside triphosphates as substrates. Responsible for the transcription of early, intermediate and late genes.	[46]
16	AAL40575.1	Q8V4W9	Virion morphogenesis protein OPG132	A6 is essential for virion membrane biogenesis and localization of virion membrane proteins to sites of virion assembly.	[59]
17	AAL40576.1	Q8V4W8	Early transcription factor 82 kDa subunit	A viral transcription factor subunit. Initiates early gene transcription.	[60]

18	AAL40577.1	Q8V4W7	Intermediate transcription factor 3 small subunit	Acts with RNA polymerase to initiate transcription from intermediate gene promoters.	[61]
19	AAL40579.1	Q8V4W5	Major core protein OPG136 precursor	Core protein 4a is the most abundant virion protein. Major component of the virion core that undergoes proteolytic processing during the immature virion (IV) to mature virion (MV) transition.	[62]
20	AAL40598.1	Q8V4U9	Envelope protein OPG155	Envelope protein required for virus entry into host cell and for cell-cell fusion.	[63], [64]
21	AAL40599.1	Q8V4U8	DNA-directed RNA polymerase 35 kDa subunit	Part of the DNA-dependent RNA polymerase which catalyzes the transcription of viral DNA into RNA using the four ribonucleoside triphosphates as substrates. Responsible for the transcription of early, intermediate and late genes.	[46]
22	AAL40590.1	Q8V4V6	A22R	A22R, required during DNA replication for cleaving viral DNA concatamers to yield unit-length viral genomes. A20R involved in DNA replication.	[65], [66]

4.6 Structural Modelling and Model Selection

Structural analysis of the shortlisted non-homologous Mpox virus proteins was performed using obtain reliable three-dimensional models for downstream structure-based studies. Among 22 proteins, the protein AAL40526.1 have experimentally resolved structures available in the Protein Data Bank (PDB), it was associated with multiple high-resolution crystal structures (PDB IDs: 9LIK, 9LIL, and 9LIM), determined using X-ray diffraction, and was therefore directly used for further analysis without the need for modelling. Among these, the structure with PDB ID 9LIK (resolution: 2.64 Å) was selected for further analysis. This structure was chosen as a representative model due to its completeness and suitability for downstream structure-based studies.

For proteins lacking experimentally determined structures, homology modelling was carried out using the SWISS-MODEL server (Figure 6). Suitable templates were identified based on sequence similarity, and three-dimensional models were generated accordingly. To

ensure structural reliability, a sequence identity threshold of $\geq 30\%$ between the target protein and template was applied. Proteins for which suitable templates did not meet this criterion were excluded, as low sequence identity can significantly compromise model accuracy. Based on this criterion, seven proteins (AAL40516.1, AAL40522.1, AAL40528.1, AAL40541.1, AAL40565.1, AAL40570.1, and AAL40572.1) were removed from further analysis due to insufficient template similarity. Following this filtering step, a total refined set of 15 structurally reliable protein models were selected for down-stream processing.

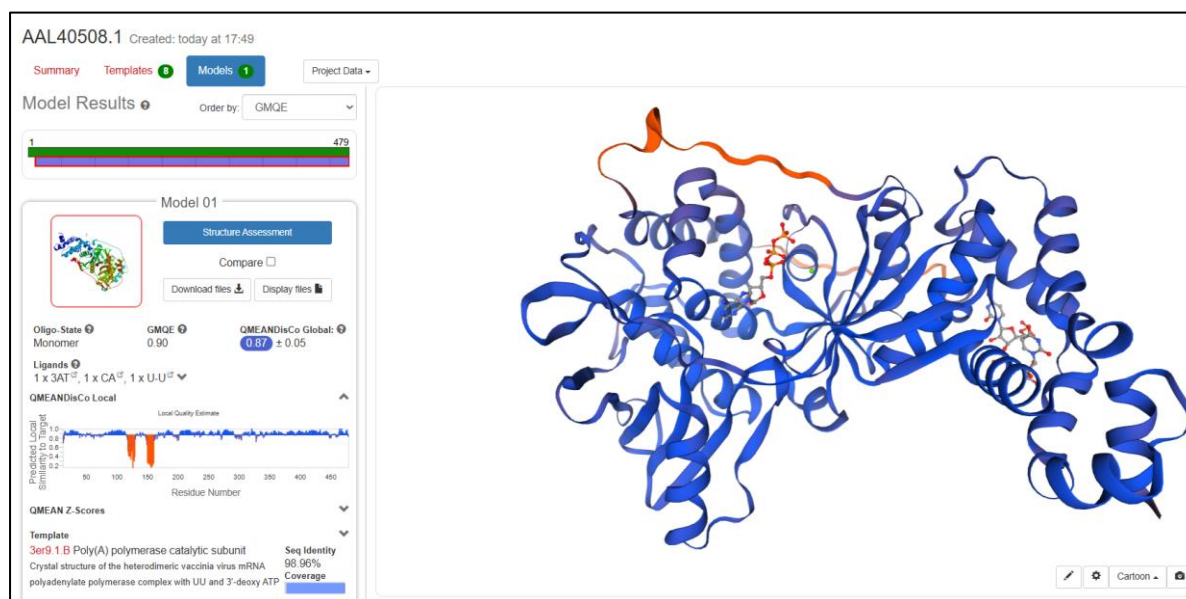


Figure 6: Structural models were generated using the SWISS-MODEL server. The highest-ranked model was selected based on GMQE and QMEAN scoring metrics, and overall model quality and local residue reliability are illustrated.

4.7 Protein Domain Identification

Following structural modelling, functional domain mapping was performed to define biologically relevant regions within each protein (Figure 7). Domain annotations were retrieved from UniProt by providing either the corresponding NCBI accession ID or UniProt ID as input. For proteins with multiple annotated features, only domains supported by Pfam were considered, ensuring reliability and consistency of domain assignments.

For each protein, domain name, start and end residue positions were extracted and systematically compiled (Table 3). Proteins lacking any annotated domain entries in UniProt were excluded from further analysis, as absence of conserved functional regions limits their suitability for downstream structure-based investigations. Out of the initial 15 proteins, one protein (AAL40574.1) did not contain any domain annotation and was therefore removed from subsequent analyses.

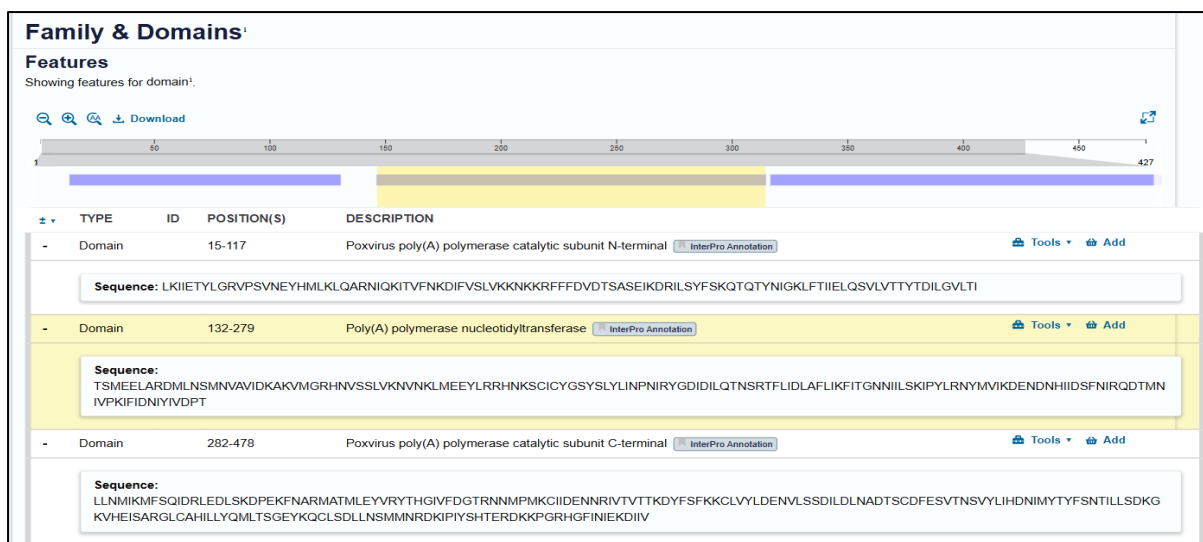


Figure 7: Domain architecture and positional annotations retrieved from UniProt, showing curated domain boundaries and feature mapping.

This domain filtering step ensured that only proteins with well-defined functional regions were retained, thereby improving the biological relevance of pocket mapping and downstream druggability assessment. These curated domain boundaries also served as a critical reference for evaluating the spatial localization of predicted binding pockets within functionally significant regions. After this step from a set of 15 proteins, only 14 proteins were left for further evaluation.

Table 3: Functional domain annotations of selected proteins obtained from UniProt, including domain names, residue boundaries, and filtering status based on domain availability.

S. No.	Protein NCBI ID	Protein Name	Domain Name	Start Residue	End Residue	Status
1	AAL40508.1	Poly(A) polymerase catalytic subunit	Poxvirus poly(A) polymerase catalytic subunit C-terminal	282	478	Retained
2	AAL40511.1	DNA-directed RNA polymerase 30 kDa polypeptide	Zinc finger TFIIIS-type profile	155	195	Retained
3	AAL40517.1	Core protein OPG073	Virion core protein, vaccinia E11L type	1	127	Retained
4	AAL40526.1	I7L	Peptidase C57, Vaccinia virus protein I7	1	420	Retained
5	AAL40550.1	H2R	Viral late protein H2	1	188	Retained
6	AAL40554.1	DNA topoisomerase I	DNA topoisomerase I catalytic core eukaryotic-type	92	277	Retained
7	AAL40564.1	E9R	Nudix hydrolase	30	209	Retained
8	AAL40575.1	Virion morphogenesis protein OPG132	Poxvirus A6	1	368	Retained

9	AAL40576.1	Early transcription factor 82 kDa subunit	Poxvirus early transcription factor, large subunit	3	702	Retained
10	AAL40577.1	Intermediate transcription factor 3 small subunit	VITF-3 subunit protein	1	287	Retained
11	AAL40579.1	Major core protein OPG136 precursor	Poxvirus P4A	12	887	Retained
12	AAL40598.1	Envelope protein OPG155	Poxvirus A28	1	146	Retained
13	AAL40599.1	DNA-directed RNA polymerase 35 kDa subunit	DNA-directed RNA polymerase, 35kDa subunit, poxviral	3	304	Retained
14	AAL40590.1	A22R	Chordopoxvirus A20R	1	332	Retained
15	AAL40574.1	DNA-directed RNA polymerase	N/A	N/A	N/A	Excluded

4.8 Structural Validation and Quality Assessment

The structural quality of the shortlisted Mpox virus proteins was evaluated using the SAVES server, incorporating multiple validation tools including ERRAT, VERIFY3D, and PROCHECK. A total of 14 protein structures were assessed to determine their suitability for downstream structure-based analyses.

The ERRAT analysis indicated that the majority of the protein models exhibited high overall quality, with several structures achieving scores above 90, suggesting reliable non-bonded atomic interactions. Notably, proteins such as Virion morphogenesis protein OPG132, DNA topoisomerase I, and E9R demonstrated particularly high ERRAT scores, reflecting strong structural integrity. VERIFY3D analysis showed variable results across the protein set. While a limited number of proteins passed the VERIFY3D threshold, a substantial proportion exhibited partial compatibility between the 3D structure and the amino acid sequence. Such variations are commonly observed in homology-based models, especially in flexible or less conserved regions. PROCHECK analysis further supported the overall structural quality, with most proteins demonstrating acceptable stereochemical properties. Several models showed low error counts and a higher number of residues in favorable regions, indicating good backbone geometry and conformational stability.

Taken together, the combined validation results suggest that the majority of the protein structures are of acceptable quality for downstream applications, despite minor deviations observed in certain validation parameters. The use of multiple validation tools provides a comprehensive assessment of structural reliability, ensuring that only structurally reasonable models are utilized for binding pocket identification and molecular docking. The complete

validation results, including ERRAT scores, VERIFY3D outcomes, and PROCHECK statistics for all proteins, are provided in Table 4.

Table 4: Structural validation summary of selected Mpox proteins evaluated using the SAVES Server.

S. No.	NCBI Accession ID	ERRAT Score	VERIFY3D Result	PROCHECK Summary
1	AAL40508.1	94.9772	Fail (61.11%)	Errors: 4, Warning: 3, Pass:2
2	AAL40511.1	68.2927	Fail (21.86%)	Errors: 4, Warning: 2, Pass:2
3	AAL40517.1	92.093	Fail (50%)	Errors: 1, Warnings: 5, Pass: 2
4	AAL40526.1	96.3871	Fail (68.8%)	Errors: 3, Warnings: 2, Pass: 3
5	AAL40550.1	91.8367	Fail (52.80%)	Errors: 0, Warnings: 5, Pass: 4
6	AAL40554.1	95.082	Pass (84.35%)	Errors: 2, Warnings: 3, Pass: 3
7	AAL40564.1	95.7895	Fail (68.10%)	Errors: 3, Warnings: 3, Pass: 2
8	AAL40575.1	97.5104	Fail (26.51%)	Errors: 1, Warnings: 4, Pass: 3
9	AAL40576.1	91.6159	Fail (65.54%)	Errors: 2, Warnings: 4, Pass: 2
10	AAL40577.1	96.0474	Fail (44.79%)	Errors: 1, Warnings: 4, Pass: 3
11	AAL40579.1	92.7252	Fail (78.85%)	Errors: 3, Warnings: 3, Pass: 3
12	AAL40598.1	88.1356	Fail (32.41%)	Errors: 0, Warnings: 4, Pass: 4
13	AAL40599.1	94.8097	Fail (69.64%)	Errors: 0, Warnings: 5, Pass: 3
14	AAL40590.1	93.7811	Fail (76.29%)	Errors: 2, Warnings: 4, Pass: 2

4.9 Binding Pocket Detection

Binding pocket analysis of the 14 selected Mpox viral proteins revealed clear variation in pocket geometry, physicochemical characteristics, and overall druggability potential. Key parameters—including pocket volume, surface area, depth and drug score—were evaluated using the ProteinPlus DoGSiteScorer tool.

Multiple binding pockets were identified for all protein structures. Pockets with drug scores above 0.7 were considered optimal for ligand binding. In cases where fewer pockets satisfied this threshold, the highest-ranking pockets were retained to ensure adequate representation for downstream analysis.

Overall, the results indicate that pocket druggability is governed by a combination of structural and chemical features rather than pocket size alone. Factors such as depth, residue composition, hydrophobicity, and hydrogen bonding capacity collectively contribute to the suitability of a binding site for ligand interaction. The predominance of well-defined and druggable pockets across the majority of proteins supports their selection for subsequent molecular docking analyses.

All detailed pocket parameters such as pocket number, pocket volume ($\text{\AA}^3$), pocket surface area ($\text{\AA}^2$), pocket drug score, depth ($\text{\AA}$), hydrogen bond donors and acceptors, number of metals in pocket, hydrophobic ratio, apolar amino acid ratio, polar amino acid ratio, positive

amino acid ratio, negative amino acid ratio and number of atoms in each pocket are provided in Supplementary Table S3. These physicochemical descriptors are used for further ligand selection.

4.10 Domain Pocket Association Mapping

To further refine the biological relevance of the predicted binding sites, pocket–domain association analysis was performed by mapping each identified pocket onto its corresponding functional domain (Figure 8). This step enabled prioritization of pockets that are not only structurally favorable but also located within biologically meaningful regions of the protein.

Overall, a strong correspondence between predicted binding pockets and annotated functional domains was observed across the dataset (Table 5). The majority of high-scoring pockets exhibited high domain association (>70%), indicating that these pockets are well-aligned with functionally important regions. Notably, several proteins—including AAL40517.1, AAL40526.1, AAL40550.1, AAL40564.1, AAL40575.1, AAL40576.1, AAL40577.1, AAL40579.1, AAL40598.1, and AAL40599.1—demonstrated complete or near-complete overlap (approaching or equal to 100%) between pocket atoms and domain residues. This strong spatial agreement suggests that the predicted pockets likely represent true functional binding sites rather than incidental surface cavities.

In particular, multiple pockets within individual proteins consistently showed high association values. For example, all identified pockets in AAL40526.1 and AAL40576.1 exhibited 100% domain overlap, reinforcing their structural and functional reliability as docking targets. Similarly, AAL40579.1 displayed consistently high association values (>95%) across multiple pockets, further supporting the robustness of these predicted binding regions. These findings indicate that the identified pockets are not randomly distributed but are concentrated within conserved and functionally significant regions.

Moderate levels of association were observed in a few cases. AAL40508.1 exhibited mixed behavior, with one pocket showing moderate association (~61%) and another demonstrating high association (~85%), suggesting partial localization within the functional domain. Likewise, AAL40590.1 showed variability, with one pocket falling within the medium association range (~68%) while others displayed complete overlap. These results indicate that while some pockets are fully embedded within domains, others may lie at domain boundaries or partially extend into adjacent regions.

In contrast, AAL40511.1 showed low domain association (~47%), indicating that a significant portion of the predicted pocket lies outside the annotated functional domain. This

reduced overlap suggests that the identified pocket may not correspond to a biologically relevant active or binding site, aligning with its previously observed lower druggability. Taken together, the results demonstrate that most predicted binding pockets are not only structurally favorable but also biologically meaningful, as evidenced by their strong localization within functional domains. This dual validation—combining druggability assessment with domain association—provides increased confidence in the selection of pockets for downstream molecular docking studies.

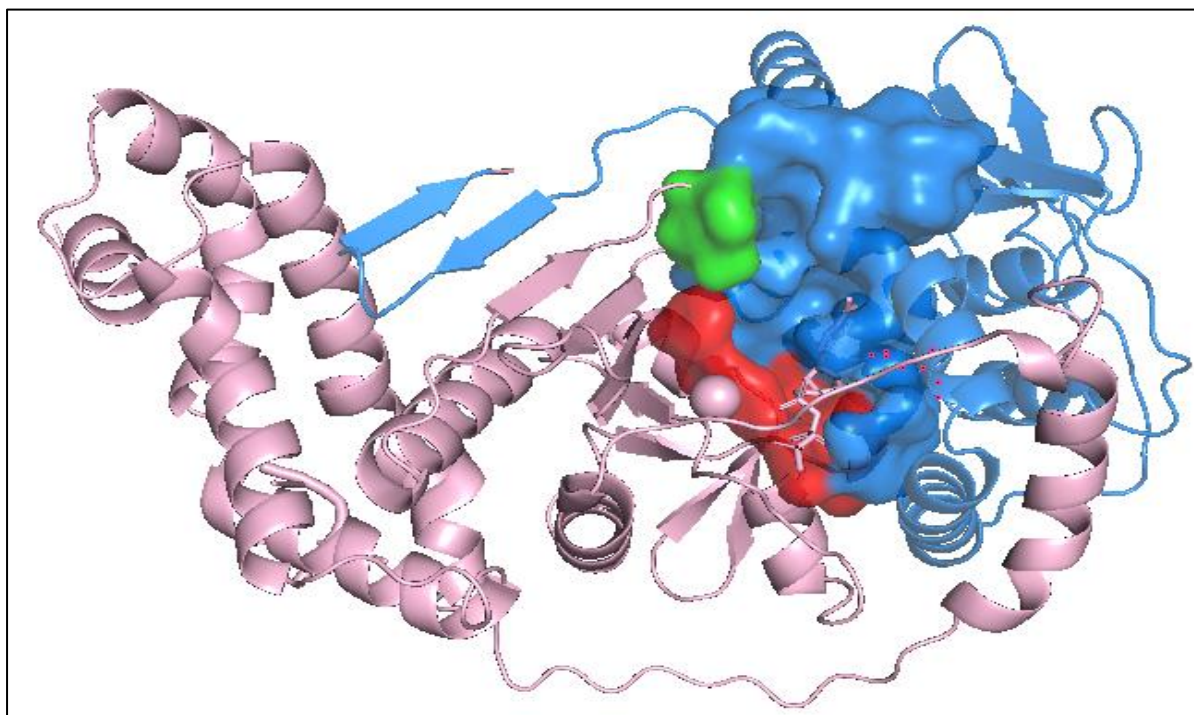


Figure 8: Visualization of predicted binding pockets within the functional domain of the Mpox viral protein Poly(A) polymerase catalytic subunit (NCBI ID: AAL40508.1). The protein structure is represented in cartoon format (pale pink), with the C-terminal catalytic domain (residues 282–478) highlighted in marine blue. The predicted binding pockets are shown as surface representations, with P₀ in red and P₁ in green, illustrating that both pockets are localized within the annotated functional domain.

Table 5: Druggability class along with protein's pocket ID and percentage of atoms associated in domain.

S. No.	NCBI Accession ID	Pocket ID	Number of Atoms in Pocket	Number of Pocket Atoms Inside Domain	Percentage of Pocket Atoms in Domain: <i>pocket atoms inside domain / total atoms in pocket * 100 = %</i>	Druggability Class
1	AAL40508.1	P_0	122	75	61.40%	Medium
		P_1	145	123	84.80%	High
2	AAL40511.1	P_0	191	91	47.60%	Low
3	AAL40517.1	P_0	84	79	94%	High
		P_1	127	127	100	High

		P_2	134	134	100	High
4	AAL40526.1	P_0	355	355	100%	High
		P_1	188	188	100%	High
		P_2	154	154	100%	High
		P_4	174	174	100%	High
5	AAL40550.1	P_0	101	101	100%	High
		P_2	62	62	100%	Medium
6	AAL40554.1	P_0	330	286	87%	High
		P_1	88	83	94.31%	High
		P_2	81	81	100%	High
7	AAL40564.1	P_0	231	208	90%	High
		P_1	118	118	100%	High
8	AAL40575.1	P_0	158	158	100%	High
		P_1	131	109	83%	High
		P_2	134	134	100%	High
		P_3	101	101	100%	High
		P_4	124	124	100%	High
9	AAL40576.1	P_0	265	244	92%	High
		P_1	143	143	100%	High
		P_2	154	154	100%	High
		P_3	138	138	100%	High
		P_4	101	101	100%	High
		P_5	117	117	100%	High
		P_8	122	122	100%	High
10	AAL40577.1	P_0	274	274	100%	High
		P_1	189	189	100%	High
		P_2	122	122	100%	High
		P_4	50	50	100%	Medium
11	AAL40579.1	P_3	161	161	100%	High
		P_4	167	167	100%	High
		P_5	186	181	97.30%	High
		P_7	138	133	96.30%	High
		P_8	138	135	97.82%	High
12	AAL40598.1	P_0	196	196	100%	High
		P_2	48	48	100%	Low
		P_5	39	39	100%	Low
13	AAL40599.1	P_0	135	135	100%	High

14	AAL40590.1	P_1	113	113	100%	Medium
		P_3	68	68	100%	Medium
		P_0	168	115	68.45%	High
		P_1	159	159	100%	High
		P_2	108	108	100%	Medium
		P_4	81	81	100%	Medium

4.11 Molecular Docking

4.11.1 Binding Pocket Refinement and Selection for Virtual Screening

During pocket screening, several predicted cavities were excluded from downstream docking analysis due to unsuitable structural and functional characteristics. Pockets exhibiting excessively large ($>1000 \text{ \AA}^3$) or very small ($<300 \text{ \AA}^3$) cavity volumes were removed, as these regions may either promote non-specific ligand interactions or provide insufficient space for stable ligand accommodation. In addition, pockets showing low druggability scores or poor overlap with annotated functional domains were not retained for further analysis.

Based on these filtering criteria, the pockets AAL40511.1 (P_0), AAL40526.1 (P_0), AAL40554.1 (P_0), AAL40550.1 (P_2), AAL40576.1 (P_0), AAL40575.1 (P_4), AAL40577.1 (P_0 and P_4), AAL40598.1 (P_2 and P_5), AAL40590.1 (P_0 and P_4), and AAL40599.1 (P_3) were eliminated from subsequent virtual screening analyses.

The remaining pockets (Table 6) exhibited suitable cavity dimensions together with favourable druggability and functional-domain association, supporting their selection as biologically relevant docking sites for molecular screening studies.

Table 6: List of retained proteins and pockets after binding pocket refinement along with their volume and druggability class.

S. No.	NCBI Accession ID	Pocket ID	Pocket Volume ($\text{\AA}^3$)	Druggability Class
1	AAL40508.1	P_0	694.89	Medium
		P_1	512.72	High
2	AAL40517.1	P_0	472.64	High
		P_1	371.01	High
		P_2	335.74	High
3	AAL40526.1	P_1	785.13	High
		P_2	610.64	High
		P_4	571.99	High
4	AAL40550.1	P_0	581.89	High
5	AAL40554.1	P_1	360.71	High

		P_2	346.92	High
6	AAL40564.1	P_0	1082.11	High
		P_1	466.37	High
7	AAL40575.1	P_0	971.39	High
		P_1	699.71	High
		P_2	698.18	High
		P_3	468.35	High
8	AAL40576.1	P_1	757.23	High
		P_2	557.34	High
		P_3	511.48	High
		P_4	505.02	High
		P_5	495.61	High
		P_8	378.42	High
9	AAL40577.1	P_1	496.21	High
		P_2	391.89	High
10	AAL40579.1	P_3	804.36	High
		P_4	739.03	High
		P_5	693.6	High
		P_7	537.65	High
		P_8	498.07	High
11	AAL40598.1	P_0	839.2	High
12	AAL40599.1	P_0	446.4	High
		P_1	329.41	Medium
13	AAL40590.1	P_1	587.66	High
		P_2	420.07	Medium

4.11.1 Ligand Selection and Structure Retrieval

Following the identification, characterization and refining of binding pockets, a total of 13 proteins and 35 pockets were retained for further downstream processing.

Ligand selection was carried out using a pocket-guided approach, integrating compound libraries from ZINC12 database and structural data in SDF format from PubChem. For each retained binding pocket, 5 candidate ligands were shortlisted, resulting in a curated ligand set optimized for structural compatibility and drug-likeness. The final dataset was organized and presented in Supplementary Table S4, where each ligand is represented by its corresponding PubChem Compound ID (CID) and SMILES (Simplified Molecular Input Line Entry System). SMILES representations were retrieved and retained, SMILES strings serve as

a standardized and computationally efficient format for molecular representation, enabling seamless integration into downstream ADMET prediction workflows. A total of 175 ligands for 35 pockets and 13 proteins were selected and retrieved.

4.11.2 Ligand Preparation

A total of 175 ligands were successfully processed and standardized to generate a uniform, docking-compatible dataset. The preparation workflow resulted in structurally consistent ligand representations, ensuring that all compounds were optimized to energetically stable conformations and encoded with appropriate atomic and charge parameters. The finalized ligand set exhibited high structural integrity and compatibility with downstream computational tools, enabling seamless integration into docking pipelines. Conversion into a standardized format further ensured uniformity across all compounds, minimizing variability arising from structural inconsistencies.

Overall, the ligand preparation phase yielded a high-quality, analysis-ready compound library, providing a robust foundation for reliable docking simulations and subsequent ADMET evaluations.

4.11.3 Protein Preparation

A total of 13 protein structures were successfully processed and standardized for downstream docking analysis. The preparation workflow resulted in structurally consistent protein models, free from non-essential components that could interfere with ligand binding.

All processed proteins exhibited clean and well-defined binding environments, enabling accurate representation of interaction sites. The removal of extraneous molecular entities ensured that the binding pockets remained unobstructed, allowing for reliable assessment of ligand accessibility and interaction potential. Furthermore, the incorporation of appropriate atomic parameters and structural refinements contributed to the generation of energetically and chemically coherent protein models. This standardization minimized structural variability across the dataset and ensured compatibility with docking algorithms.

Overall, the prepared protein set represents a high-quality and uniform structural framework, providing a stable and reliable foundation for subsequent docking simulations and interaction analysis.

4.11.4 Docking Grid Implementation Across Protein Targets

Docking grid parameters varied across the selected protein pockets according to the spatial architecture and geometric extent of each predicted binding cavity. Grid centers and box

dimensions were individually optimized based on pocket-specific minimum and maximum coordinates obtained from PyMOL, ensuring accurate coverage of the functional binding regions while minimizing unnecessary search space. The resulting grid dimensions ranged from compact cavities to relatively broader binding environments, reflecting the structural diversity observed among the retained Mpox protein targets.

Proteins containing multiple predicted pockets exhibited notable variation in grid box dimensions and spatial positioning, indicating the presence of structurally distinct ligand-binding regions within the same protein architecture. This pocket-specific grid optimization enabled focused docking simulations tailored to the physicochemical and geometric properties of each retained cavity.

Following pocket refinement and volume-based filtering, docking simulations were performed only on structurally relevant pockets exhibiting favourable druggability characteristics. The selected grid configurations provided sufficient conformational space for ligand accommodation and exploration, while maintaining localization within biologically relevant functional domains.

Table 7: Grid box parameters and spatial coordinates of selected binding pockets used for molecular docking of prioritized Mpox protein targets.

S. No.	NCBI Accession ID	Pocket ID	Pocket Minimum and Maximum Extent	Pocket Grid Centre (X, Y, Z) $X_c = (X_{min} + X_{max})/2$	Grid Size / Dimensions (Å) (X, Y, Z)
1	AAL40508.1	P_0	Min: [10.039, -1.263, 6.477] ; Max: [27.010, 20.483, 26.100]	[18.525, 9.610, 16.289]	22 * 27 * 25
		P_1	Min: [3.615, -2.413, -3.277] ; Max: [19.387, 20.451, 14.637]	[11.501, 9.019, 5.680]	22 * 29 * 24
2	AAL40517.1	P_1	Min: [-31.328, -46.052, 29.252] ; Max: [-10.043, -27.593, 51.554]	[-20.686, -36.823, 40.403]	27 * 24 * 28
		P_0	Min: [-44.875, -41.593, 23.031] ; Max: [-29.885, -22.476, 37.375]	[-37.380, -32.035, 30.203]	21 * 25 * 20
		P_2	Min: [-21.332, -32.640, 26.037] ; Max: [-6.528, -18.140, 52.612]	[-13.930, -25.390, 39.325]	21 * 21 * 33
3	AAL40526.1	P_1	Min: [65.294, 54.026, -30.775] ; Max: [90.923, 70.025, -5.459]	[78.109, 62.026, -18.117]	32 * 22 * 31

		P_2	Min: [9.299, 50.333, 14.498] ; Max: [30.051, 70.159, 3.547]	[19.675, 60.246, 24.023]	27 * 26 * 25
		P_4	Min: [3.216, 44.929, -0.299] ; Max: [30.894, 57.911, 2.4207]	[17.055, 51.420, 11.954]	34 * 19 * 31
4	AAL40550.1	P_0	Min: [166.096, 153.269, 147.170] ; Max: [187.816, 168.593, 164.259]	[176.956, 160.931, 155.715]	28 * 22 * 23
5	AAL40554.1	P_1	Min: [24.118, -16.424, 35.794] ; Max: [41.110, -3.137, 53.102]	[32.614, -9.781, 44.448]	23 * 19 * 23
		P_2	Min: [3.524, -24.033, -1.666] ; Max: [18.193, -3.958, 11.125]	[10.859, -13.996, 4.730]	21 * 26 * 19
6	AAL40564.1	P_0	Min: [-32.991, -35.857, -13.511] ; Max: [-2.770, -17.352, 8.247]	[-17.881, -26.605, -2.632]	36 * 25 * 28
		P_1	Min: [-28.883, -36.913, -31.886] ; Max: [-17.046, -15.486, -6.736]	[-22.965, -26.200, -19.311]	16*25*29
7	AAL40575.1	P_0	Min: [-47.638, 12.973, -62.284] ; Max: [-17.712, 35.583, -39.057]	[-32.675, 24.278, -50.671]	30 * 24 * 24
		P_1	Min: [-46.031, 2.246, -75.360] ; Max: [-22.877, 19.123, -53.653]	[-34.454, 10.685, -64.507]	29 * 23 * 28
		P_2	Min: [-34.268, 9.977, -74.109] ; Max: [-11.488, 29.315, -45.842]	[-22.878, 19.646, -59.976]	29 * 25 * 34
		P_3	Min: [-54.676, -1.487, -65.563] ; Max: [-42.159, 22.865, -50.269]	[-48.418, 10.689, -57.916]	19 * 30 * 21
8	AAL40576.1	P_1	Min: [-65.491, 7.254, -18.163] ; Max: [-47.446, 26.864, 11.263]	[-56.469, 17.059, -3.450]	24 * 26 * 35
		P_2	Min: [-28.540, 37.351, -30.535] ; Max: [-11.542, 56.879, -10.988]	[-20.041, 47.115, -20.762]	23 * 26 * 26

		P_3	Min: [-11.478, 30.707, -42.650] ; Max: [6.858, 46.825, -19.250]	[-2.310, 38.766, -30.950]	25 * 22 * 30
		P_4	Min: [-68.811, 20.917, -25.365] ; Max: [-47.594, 34.958, -8.364]	[-58.203, 27.938, -16.865]	27 * 20 * 23
		P_5	Min: [-38.014, 6.905, 17.780] ; Max: [-18.589, 25.076, 38.138]	[-28.302, 15.991, 27.959]	26 * 24 * 27
		P_8	Min: [-38.874, 23.856, -33.328] ; Max: [-22.825, 41.905, -13.993]	[-30.850, 32.881, -23.661]	22 * 24 * 25
9	AAL40577.1	P_1	Min: [180.503, 163.053, 221.582] ; Max: [206.471, 184.935, 236.912]	[193.487, 173.994, 229.247]	32 * 28 * 21
		P_2	Min: [197.223, 156.597, 211.794] ; Max: [213.522, 174.174, 227.126]	[205.373, 165.386, 219.460]	22 * 24 * 21
10	AAL40579.1	P_3	Min: [176.549, 159.745, 154.779] ; Max: [196.830, 178.190, 174.993]	[186.690, 168.968, 164.886]	26 * 25 * 26
		P_4	Min: [152.098, 141.282, 147.713] ; Max: [173.586, 161.393, 173.371]	[162.842, 151.338, 160.542]	27 * 26 * 32
		P_5	Min: [149.446, 172.179, 175.354] ; Max: [170.701, 199.024, 201.991]	[160.074, 185.602, 188.673]	27 * 33 * 33
		P_7	Min: [176.744, 163.143, 180.337] ; Max: [193.132, 182.977, 201.990]	[184.938, 173.060, 191.164]	20 * 22 * 24
		P_8	Min: [155.961, 147.756, 180.311] ; Max: [173.956, 162.842, 202.229]	[164.959, 155.299, 191.270]	24 * 21 * 28
11	AAL40598.1	P_0	Min: [187.423, 164.210, 125.227] ; Max: [212.064, 185.380, 151.850]	[199.744, 174.795, 138.539]	28 * 24 * 30
12	AAL40599.1	P_0	Min: [6.699, -48.048, -55.760] ; Max: [20.907, -23.025, -39.215]	[13.803, -35.537, -47.488]	20 * 31 * 23
		P_1	Min: [11.888, -30.887, -59.321] ; Max: [28.325, -11.384, -43.454]	[20.107, -21.136, -51.388]	22 * 26 * 22

13	AAL40590.1	P_1	Min: [101.052, 91.112, 113.383] ; Max: [119.133, 115.802, 132.486]	[110.093, 103.457, 122.935]	24 * 31 * 25
		P_2	Min: [105.456, 109.438, 102.728] ; Max: [121.707, 127.166, 117.547]	[113.582, 118.302, 110.138]	22 * 24 * 21

4.11.5 Molecular Docking Results

Molecular docking analysis identified several ligands–protein complexes with favourable binding affinities across the selected Mpox protein targets. Docking scores varied considerably between different proteins and binding pockets, reflecting differences in pocket geometry and ligand compatibility. Multiple high-affinity interactions were observed for proteins AAL40508.1, AAL40517.1, AAL40575.1, AAL40576.1, and AAL40579.1, indicating strong ligand accommodation within their predicted functional pockets.

Several ligands produced moderate or weak binding affinities and were therefore not considered for further prioritization. In addition, a small number of ligands were unable to produce stable docking conformations within certain pockets, suggesting limited structural compatibility or insufficient interaction within the defined binding cavity. Overall, the docking workflow enabled efficient screening and prioritization of structurally favourable ligand–protein interactions for subsequent pharmacokinetic and toxicity assessment.

A total of 175 ligands were initially screened during molecular docking. Following removal of the three non-docking ligands, 172 successfully docked ligands were retained and advanced to subsequent ADMET screening and pharmacokinetic evaluation. Detailed docking results of all 172 ligands, including grid dimensions, ligand PubChem IDs, and binding affinities, are provided in Supplementary Table S5.

4.12 *In-Silico* ADMET Profiling and Pharmacokinetic Risk Assessment

Ligands exhibiting docking affinities ≤ -8.0 kcal/mol were selected for downstream ADMET evaluation. The threshold of -8.0 kcal/mol was adopted because binding energies within this range are commonly considered indicative of strong and potentially stable protein–ligand interactions in virtual screening studies. Ligands with higher binding energies generally represent weaker interactions and lower binding stability, whereas compounds exhibiting affinities ≤ -8.0 kcal/mol are more likely to demonstrate biologically relevant target engagement. The SMILES representations obtained during the ligand selection stage were used as input for SwissADME and pkCSM analyses to assess physicochemical suitability, pharmacokinetic behaviour, and potential toxicity risks.

A total of 172 ligands were initially screened against 13 proteins comprising 35 selected binding pockets. Following molecular docking, only 36 ligands interacting with 5 proteins across 13 binding pockets satisfied the predefined docking threshold of ≤ -8.0 kcal/mol and were therefore subjected to ADMET profiling. Detailed binding affinities, SMILES, physicochemical, pharmacokinetic, toxicity, ADMET description, and ADMET risk categorization data for all screened ligands are provided in Supplementary Table S6. Among these, ligands categorized under low- and moderate-risk groups were retained for further prioritization, while compounds classified as moderate-high or high risk were excluded from subsequent analysis.

Overall, the shortlisted ligands demonstrated favourable drug-like characteristics, including acceptable molecular weight, suitable lipophilicity, and optimal topological polar surface area values, supporting their potential oral bioavailability. The use of advanced filtering criteria during ZINC12 compound selection contributed to the enrichment of chemically feasible and pharmacokinetically suitable molecules prior to docking.

SwissADME predictions indicated that several ligands possessed high gastrointestinal absorption and acceptable bioavailability profiles, while maintaining manageable cytochrome P450 interaction patterns. In addition, PAINS screening did not indicate major assay-interfering characteristics among the retained compounds, further supporting their suitability for downstream prioritization.

pkCSM-based toxicity assessment identified a subset of ligands with predicted hepatotoxicity and hERG II positivity. However, these predictions are derived from QSAR-based computational models and therefore represent statistical toxicity probabilities rather than experimentally confirmed toxic effects. Consequently, in vitro cytotoxicity studies, metabolic stability assays, and in vivo validation remain necessary before establishing definitive safety profiles.

ADMET risk categorization was performed by integrating pharmacokinetic and toxicity parameters obtained from SwissADME and pkCSM. Compounds exhibiting favourable physicochemical properties, high GI absorption, minimal CYP inhibition, absence of major cardiotoxicity predictions, negative AMES toxicity, and acceptable clearance profiles were categorized as low risk. Ligands showing manageable limitations, such as limited CYP interactions or predicted hepatotoxicity without severe cardiotoxicity alerts, were categorized as moderate risk. In contrast, compounds exhibiting multiple toxicity-associated indicators, including extensive CYP inhibition, AMES positivity, hepatotoxicity, hERG-associated

cardiotoxicity, or unfavourable clearance behaviour, were classified as moderate–high or high risk.

After integrated ADMET-based filtering, 11 ligand candidates associated with 5 proteins and 8 binding pockets were retained within the low- and moderate-risk categories and prioritized for downstream refinement and cross-validation studies, while compounds classified as moderate–high or high risk were excluded due to multiple toxicity alerts, unfavourable metabolic interaction profiles, or predicted cardiotoxicity concerns. The finalized ligand candidates, summarized in Supplementary Table S7, demonstrated strong binding affinities (≤ -8.0 kcal/mol), favourable physicochemical and pharmacokinetic characteristics, high predicted gastrointestinal absorption, and absence of AMES mutagenicity. Although several compounds exhibited predicted hepatotoxicity and hERG II positivity, none showed severe cardiotoxicity risk associated with hERG I inhibition. Collectively, these findings indicate a balanced combination of binding stability, drug-like behaviour, and manageable toxicity profiles, supporting the suitability of the shortlisted complexes for subsequent redocking, sensitivity analysis, and interaction validation studies.

Table 8: Proteins, binding pockets and ligands with acceptable ADMET properties along with their binding affinity (kcal/mol), ADMET description and ADMET flag.

S. No.	NCBI Accession ID	Pocket ID	PubChem CID	Binding Affinity (kcal/mol)	Description	ADMET Flag
1	AAL40508.1	P_0	3131912	-10.03	Possible drug-drug interaction, liver toxicity, moderate clearance	Moderate
		P_1	2820937	-8.099	Predicted hepatotoxicity with favourable physicochemical and pharmacokinetic properties, requires in-vitro validation	Moderate
			6986084	-8.736	Mild cardiotoxicity, risk of hepatotoxicity	Moderate
2	AAL40517.1	P_1	2368032	-8.08	Mild cardiotoxicity, risk of hepatotoxicity	Moderate
		P_0	46360552	-9.066	Mild cardiotoxicity, risk of hepatotoxicity and medium drug-drug interaction	Moderate
			26841348	-8.059	Predicted hepatotoxicity, ideal pharmacokinetics	Moderate
3	AAL40575.1	P_1	3410901	-8.039	Mild cardiotoxicity, risk of hepatotoxicity	Moderate
		P_2	11842140	-8.054	Moderate drug-drug interaction with predicted hepatotoxicity	Moderate
4	AAL40576.1	P_4	16021336	-8.582	High drug-drug interactions	Moderate
			21005142	-8.907	Mild cardiotoxicity, risk of hepatotoxicity	Moderate
5	AAL40579.1	P_4	714070	-8.4	Predicted hepatotoxicity	Moderate

4.13 Redocking Validation and Sensitivity Checks

A total of 11 ligand candidates associated with 5 proteins and 8 binding pockets were subjected to redocking analysis using refined docking parameters to evaluate binding consistency and improve pose reliability. Redocking was performed with a reduced grid dimension of $20 \times 20 \times 20$ Å and an increased exhaustiveness value of 32, enabling deeper conformational sampling within the predicted active sites.

Overall, most ligand–protein complexes retained binding affinities comparable to the initial docking results, indicating stable and reproducible interactions within the refined binding cavities. Several ligands exhibited only minor variations in docking energy following parameter refinement, supporting the robustness of the initial virtual screening workflow.

Among the shortlisted complexes, PubChem CID 3131912 interacting with protein AAL40508.1 at pocket P_0 retained the strongest and completely unchanged binding affinity of -10.03 kcal/mol after redocking, indicating highly stable pocket localization. Similarly, ligands 2368032, 46360552, 26841348, 3410901, 16021336, 21005142, and 714070 maintained docking affinities ≤ -8.0 kcal/mol following refinement, demonstrating consistent binding behaviour under stricter docking conditions.

In contrast, ligands 2820937 and 11842140 showed reduced binding affinities of -7.584 kcal/mol and -7.263 kcal/mol, respectively, after redocking and were therefore excluded from further prioritization. Only ligand–protein complexes retaining docking affinities ≤ -8.0 kcal/mol after refinement were carried forward for downstream analysis, resulting in a final set of 9 ligands associated with 5 proteins and 8 binding pockets.

The observed reduction in affinity for ligands 2820937 and 11842140 suggests that the initial larger grid configuration may have permitted energetically favourable but less localized binding conformations outside the core functional cavity.

Collectively, the redocking and sensitivity analysis confirmed the stability of the majority of prioritized ligand–protein interactions and demonstrated that the selected complexes were robust to variations in docking parameters. Detailed redocking parameters and refined docking scores are provided in Table 9.

Table 9: Comparison of initial docking and refined redocking binding affinities of prioritized ligand–protein complexes following sensitivity analysis using optimized grid parameters and increased exhaustiveness.

S. No.	NCBI Accession ID	Pocket ID	PubChem CID	Binding Affinities (kcal/mol)	Redock Binding Affinities (kcal/mol)	Grid Centre
1	AAL40508.1	P_0	3131912	-10.03	-10.03	X 18.525 Y 9.61 Z 16.289

		P_1	2820937	-8.099	-7.584	X 11.501 Y 9.019 Z 5.68
			6986084	-8.736	-8.641	X 11.501 Y 9.019 Z 5.68
2	AAL40517.1	P_1	2368032	-8.08	-8.105	X -20.686 Y -36.823 Z 40.403
		P_0	46360552	-9.066	-8.742	X -37.38 Y -32.035 Z 30.203
			26841348	-8.059	-8.07	X -37.38 Y -32.035 Z 30.203
3	AAL40575.1	P_1	3410901	-8.039	-8.601	X -34.454 Y 10.685 Z -64.507
		P_2	11842140	-8.054	-7.263	X -22.878 Y 19.646 Z -59.976
4	AAL40576.1	P_4	16021336	-8.582	-8.584	X -58.203 Y 27.938 Z -16.865
			21005142	-8.907	-8.934	X -58.203 Y 27.938 Z -16.865
5	AAL40579.1	P_4	714070	-8.4	-8.381	X 162.842 Y 151.338 Z 160.542

4.14 Frontier Molecular Orbital (FMO) Analysis and Quantum Chemical Characterization

Frontier Molecular Orbital analysis was performed to evaluate the electronic stability, chemical reactivity, and charge transfer characteristics of the shortlisted ligands. The calculated HOMO, LUMO, energy gap (ΔE), chemical hardness, softness, electronegativity, and electrophilicity values revealed notable differences in the electronic behaviour of the candidate compounds.

Among the 9 analysed ligands (Table 10), PubChem CID 3131912 exhibited the lowest HOMO–LUMO energy gap (3.44 eV) together with the highest softness value (0.58 eV⁻¹), indicating comparatively higher chemical reactivity and improved adaptability within the protein binding environment. Similarly, PubChem CID 2368032 demonstrated a low energy gap of 3.67 eV and favourable softness (0.54 eV⁻¹), suggesting efficient electronic interaction potential with target residues. PubChem CID 3410901 also showed a relatively balanced electronic profile, with a moderate ΔE value of 3.98 eV and suitable softness characteristics, supporting its potential binding stability and molecular flexibility.

In contrast, ligands such as 6986084 and 714070 displayed comparatively larger HOMO–LUMO gaps of 5.33 eV and 5.15 eV, respectively, indicating greater electronic stability but lower chemical reactivity. PubChem CID 6986084 additionally exhibited the highest electronegativity (4.16 eV) and electrophilicity index (3.25 eV), suggesting strong electron-accepting behaviour that may contribute to excessive electronic rigidity and reduced adaptability within the binding cavity.

PubChem CIDs 46360552 and 21005142 demonstrated intermediate electronic properties, including moderate energy gaps and balanced hardness–softness values, indicating

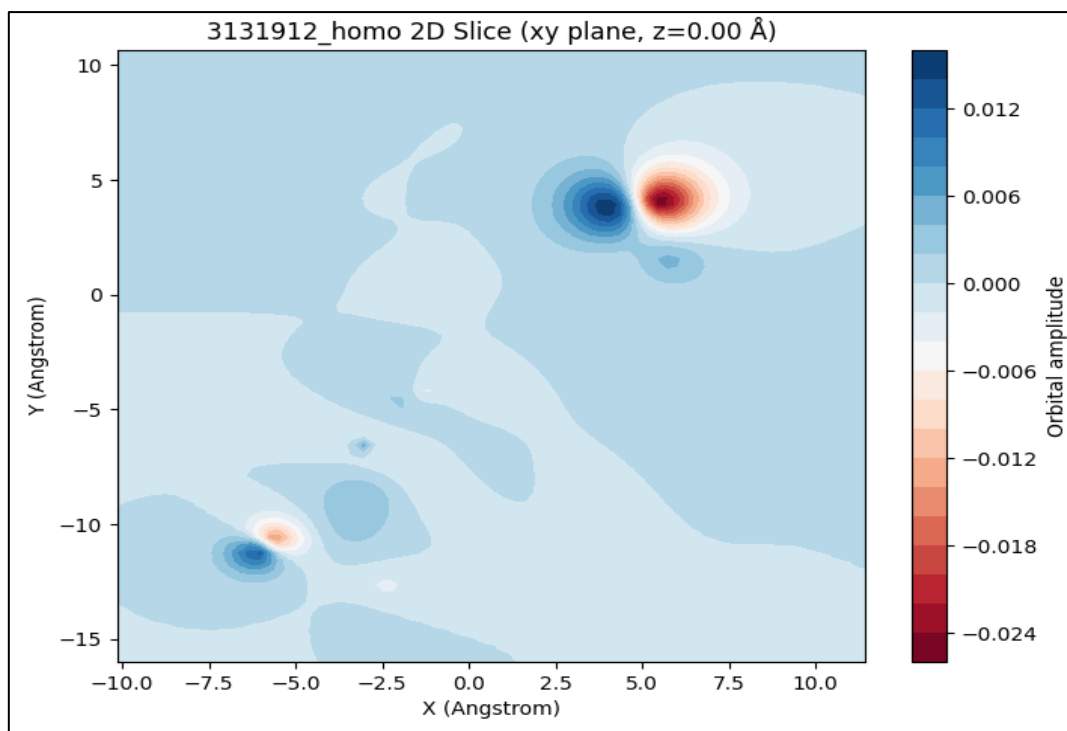
acceptable stability while maintaining sufficient reactivity for protein interaction. These ligands were therefore considered secondary favourable candidates during prioritization.

Based on the combined assessment of HOMO–LUMO energy gap, softness, electrophilicity, and overall electronic stability, ligands 3131912, 3410901, and 2368032 were identified as the most favourable compounds for downstream interaction analysis. Ligands 46360552 and 21005142 were retained as secondary prioritized candidates due to their balanced quantum chemical characteristics and stable electronic profiles.

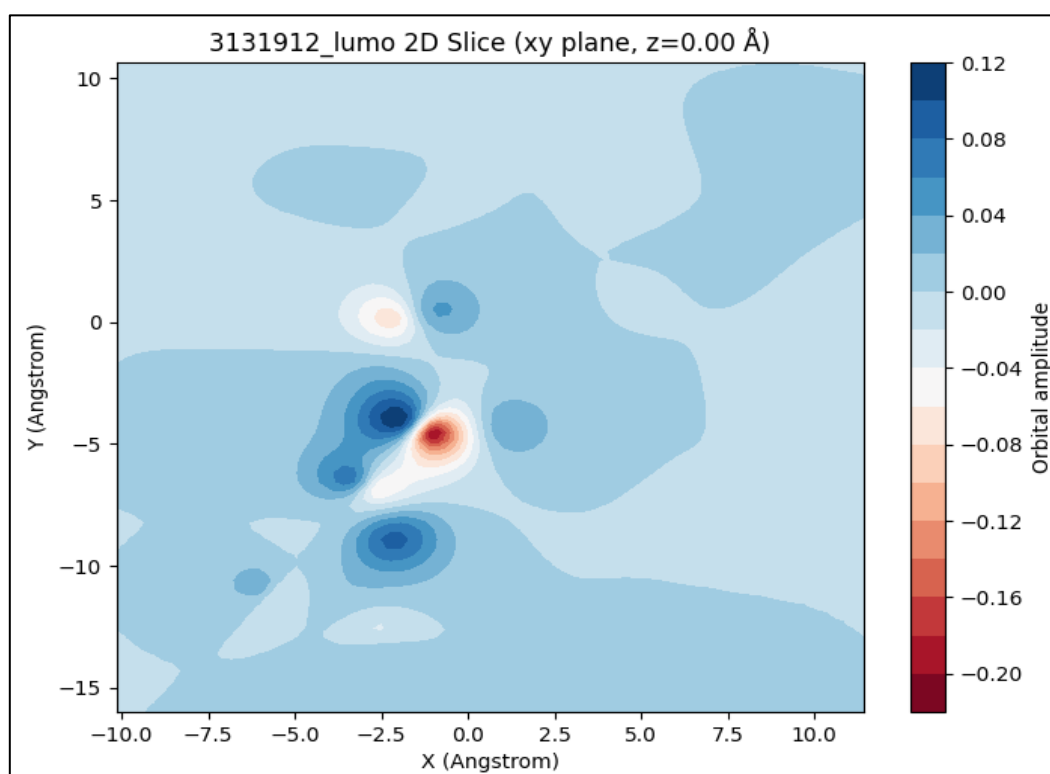
Conversely, ligands 6986084, 26841348, 16021336, 2820937, 11842140 and 714070 were excluded from further interaction analysis owing to comparatively higher energy gaps, lower softness values, or less favourable electronic reactivity profiles. Overall, the FMO analysis supported the prioritization of PubChem CIDs 3131912, 3410901, 2368032, 46360552, and 21005142 for subsequent protein–ligand interaction and residue-level binding analysis.

Table 10: Frontier Molecular Orbital (FMO)-derived quantum chemical descriptors of prioritized ligands, including HOMO and LUMO energies, energy gap (ΔE), chemical hardness, softness, electronegativity, and electrophilicity calculated using DFT-based analysis.

Pubchem ID	Homo eV	Lumo eV	Gap (ΔE) eV	Chemical Hardness (η) eV	Chemical Softness (S) eV ⁻¹	Electronegativity (χ) eV	Electrophilicity (ω) eV
3131912	-3.27	0.17	3.44	1.72	0.58	1.55	0.70
6986084	-6.83	-1.49	5.33	2.67	0.38	4.16	3.25
46360552	-3.41	1.25	4.66	2.33	0.43	1.08	0.25
2368032	-3.56	0.12	3.67	1.84	0.54	1.72	0.81
26841348	-3.90	0.88	4.78	2.39	0.42	1.51	0.48
3410901	-3.39	0.59	3.98	1.99	0.50	1.40	0.49
16021336	-2.89	1.72	4.60	2.30	0.43	0.59	0.07
21005142	-3.34	1.43	4.77	2.38	0.42	0.96	0.19
714070	-3.92	1.23	5.15	2.58	0.39	1.35	0.35



(A)



(B)

Figure 9: Two-dimensional Frontier Molecular Orbital (FMO) contour plots of PubChem CID 3131912 generated using DFT calculations. The HOMO distribution (A) and LUMO distribution (B) are shown on the xy-plane at $z = 0.00$ Å. Blue and red contour regions represent opposite orbital phases, illustrating the spatial localization of electron-donating and electron-accepting regions within the ligand.

4.15 Protein–Ligand Interaction Analysis of Prioritized Complexes

Final short-listed protein-ligand interactions, ligand 2d images, 2d interaction images and protein, active domain, ligand docked complexes are presented below.

	Protein NCBI Accession ID	Protein Name	Protein Function
1)	AAL40508.1	Poly(A) polymerase catalytic subunit	Polymerase that creates the 3'-poly(A) tail of mRNAs.

Pocket ID: P_0

PubChem CID: 3131912

Binding Affinity (kcal/mol): -10.3

ADMET Flag: Moderate

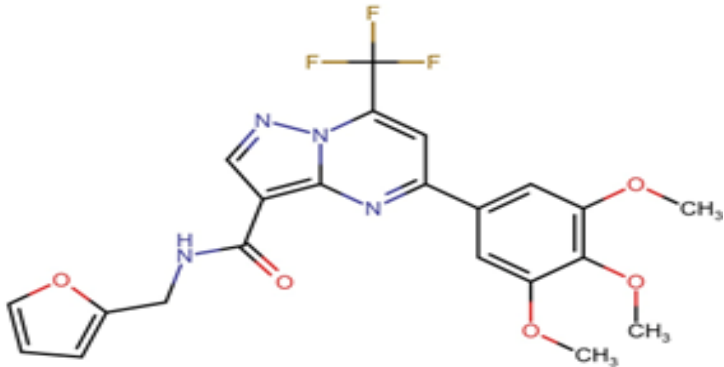
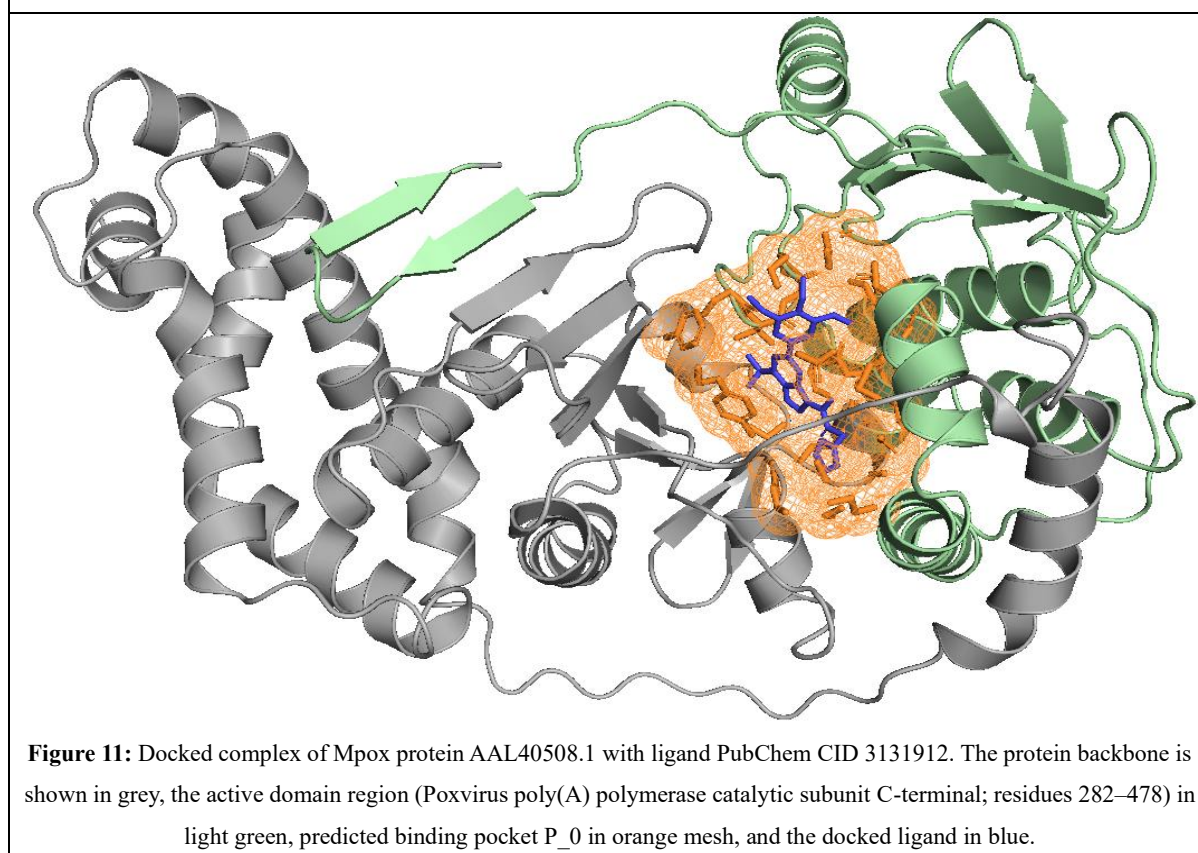
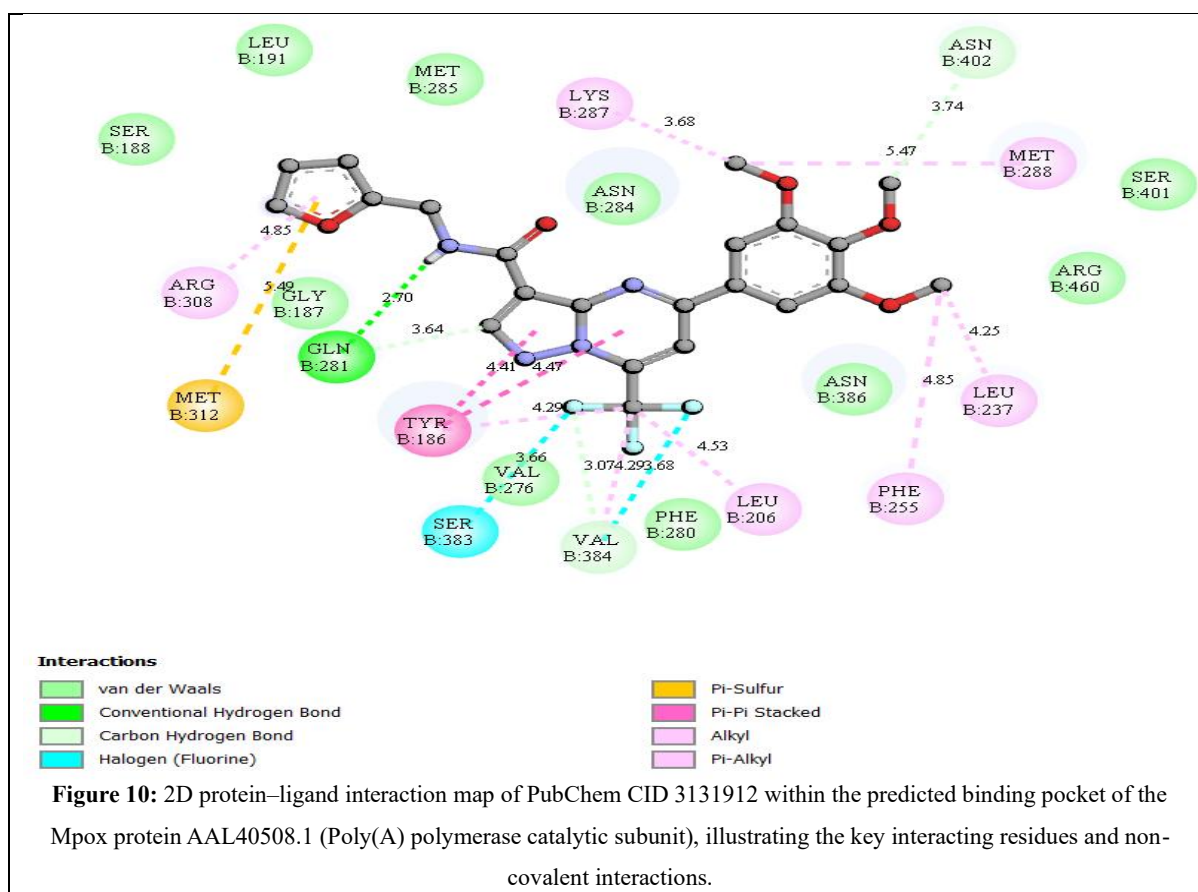


Table 11: Detailed intermolecular interaction profile of ligand 3131912 docked with protein AAL40508.1, including interaction distances (Å), bond categories, and associated van der Waals contacts within the predicted binding pocket.

Name	Distance (Å)	Category	Van der Waal's Interactions (VdW)
3131912:C - B:LEU237	4.25	Hydrophobic	LeuB:187
3131912:H - B:GLN281:OE1	2.7	Hydrogen Bond	SerB:188
B:VAL384:3131912:F	3.07	Hydrogen Bond;Halogen	LeuB:191
3131912:C - B:MET288	5.47	Hydrophobic	ValB:276
3131912 - B:TYR186	4.41	Hydrophobic	PheB:280
3131912:C - B:ASN402:OD1	3.74	Hydrogen Bond	AsnB:284
31319121:C - B:GLN281:OE1	3.64	Hydrogen Bond	MetB:285
31319121:C - B:LYS287	3.68	Hydrophobic	ValB:384
B:TYR186 - 3131912	4.47	Hydrophobic	AsnB:386
B:VAL384:O - 3131912:F	3.68	Halogen	SerB:401
B:SER383:O - 3131912:F	3.66	Halogen	AsnB:402
B:MET312:SD - 3131912	5.49	Other	ArgB:460
3131912:C - B:LEU206	4.53	Hydrophobic	
3131912:C - B:VAL384	4.29	Hydrophobic	
B:TYR186 - 3131912:C	4.29	Hydrophobic	
B:PHE255 - 3131912:C	4.85	Hydrophobic	
3131912 - B:ARG308	4.85	Hydrophobic	

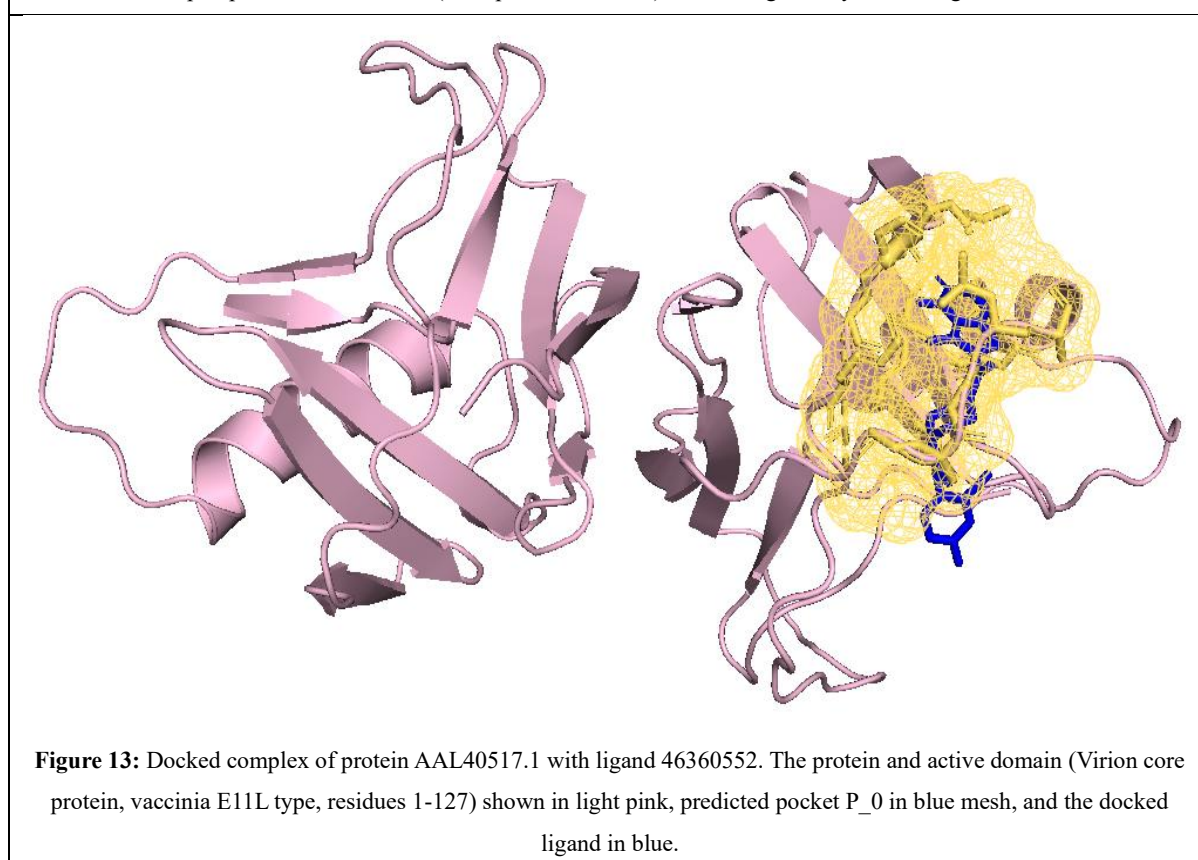
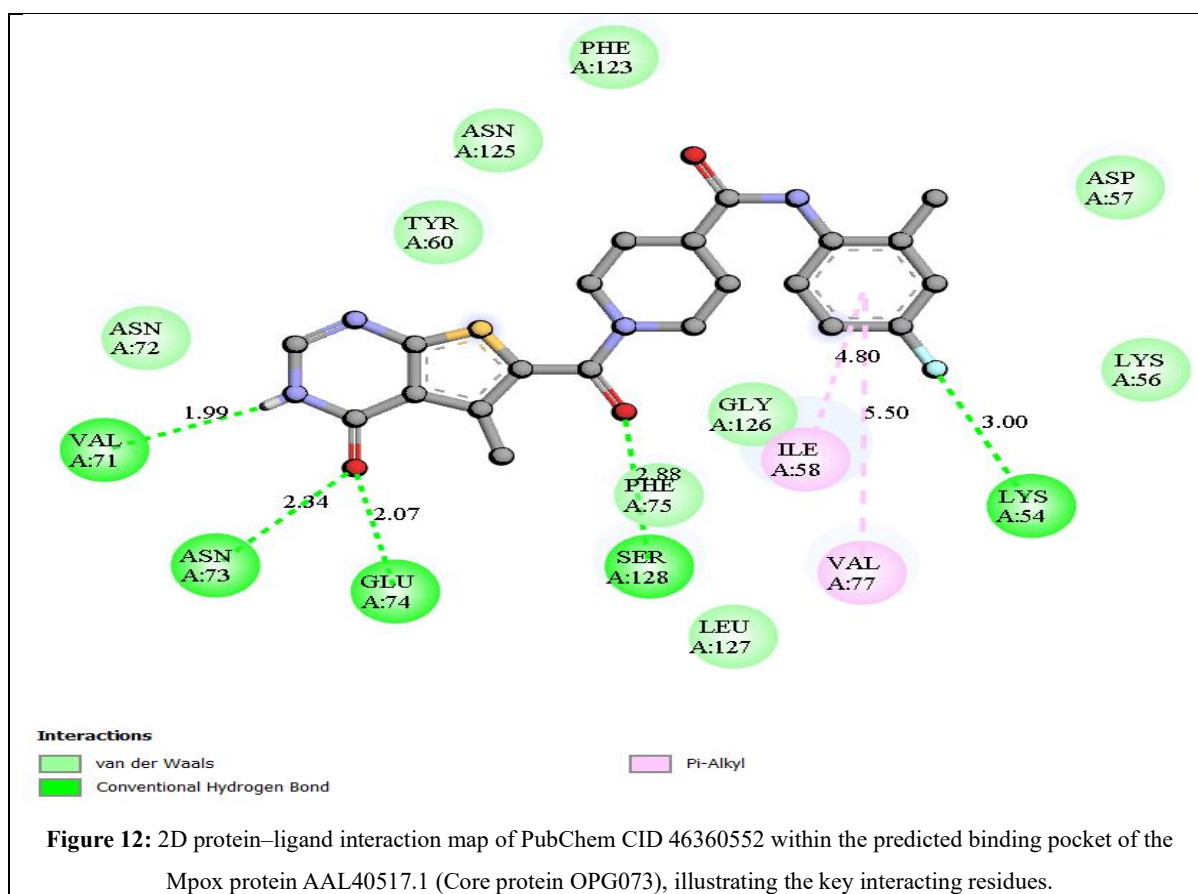


	Protein NCBI Accession ID	Protein Name	Protein Function
2)	AAL40517.1	Core protein OPG073	Virion morphogenesis

Pocket ID: P_0
PubChem CID: 46360552
Binding Affinity (kcal/mol): -8.742
ADMET Flag: Moderate

Table 12: Detailed intermolecular interaction profile of ligand 46360552 docked with protein AAL40517.1, including interaction distances (Å), bond categories, and associated van der Waals contacts within the predicted binding pocket.

Name	Distance (Å)	Category	Van der Waal's Interactions (VdW)
46360552 - A:ILE58	4.8	Hydrophobic	LysA:56
46360552:H - A:VAL71:O	1.99	Hydrogen Bond	AspA:57
A:ASN73:HN - 46360552:O	2.34	Hydrogen Bond	TyrA:60
A:LYS54:HZ1 - :46360552F	3	Hydrogen Bond;Halogen	AsnA:72
A:SER128:HN - 46360552:O	2.88	Hydrogen Bond	PheA:75
A:GLU74:HN - 46360552:O	2.07	Hydrogen Bond	PheA:123
46360552 - A:VAL77	5.5	Hydrophobic	AsnA:125
			GlyA:126
			LeuA:127



Pocket ID: P_1

PubChem CID: 2368032

Binding Affinity (kcal/mol): -8.105

ADMET Flag: Moderate

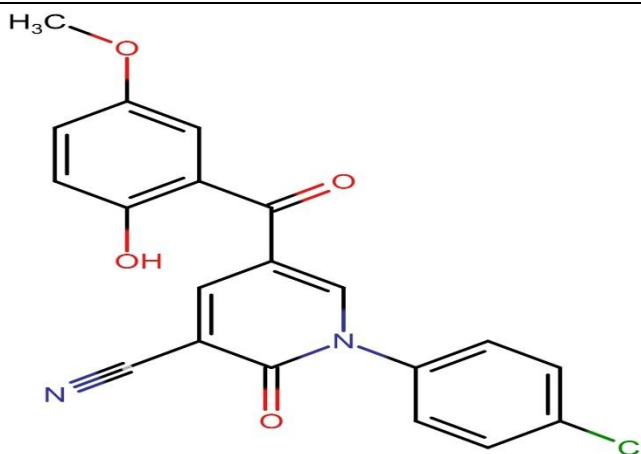
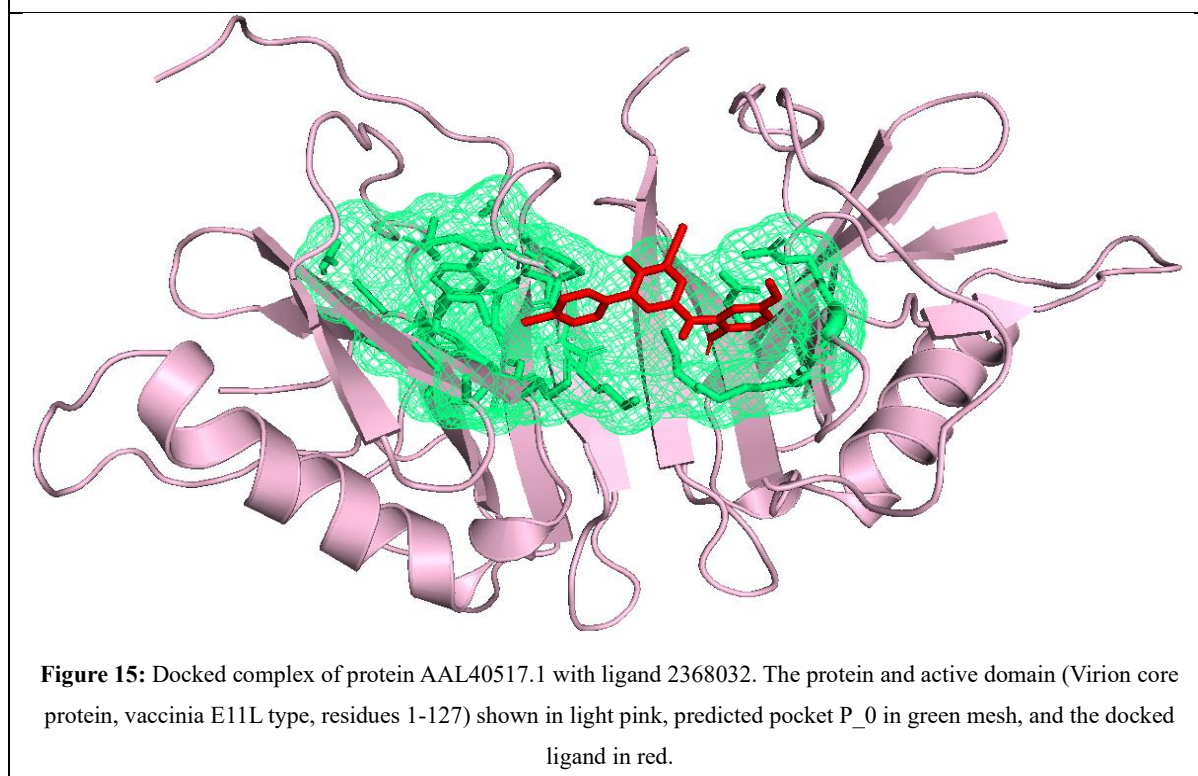
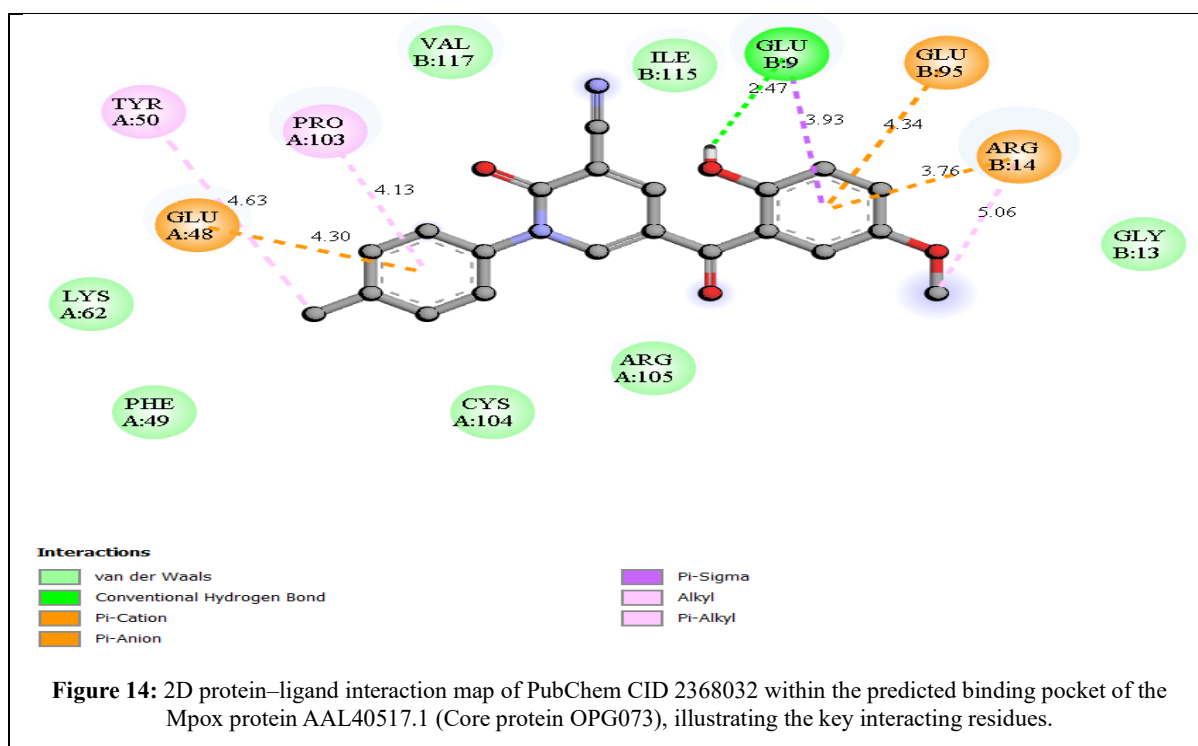


Table 13: Detailed intermolecular interaction profile of ligand 2368032 docked with protein AAL40517.1, including interaction distances (Å), bond categories, and associated van der Waals contacts within the predicted binding pocket.

Name	Distance (Å)	Category	Van der Waal's Interactions (VdW)
A:GLU48:OE1 - 2368032	4.3	Electrostatic	GlyB:13
2368032:C - B:ARG14	5.06	Hydrophobic	PheA:49
A:TYR50 - 2368032:CL	4.63	Hydrophobic	LysA:62
B:GLU9:CG - 2368032	3.93	Hydrophobic	CysA:104
B:GLU95:OE2 - 2368032	4.34	Electrostatic	ArgA:105
B:ARG14:NH2 - 2368032	3.76	Electrostatic	IleB:115
2368032:H - B:GLU9:OE2	2.47	Hydrogen Bond	ValB:117
2368032 - A:PRO103	4.13	Hydrophobic	
2368032 - B:ARG14	4.78	Hydrophobic	



	Protein NCBI Accession ID	Protein Name	Protein Function
3)	AAL40575.1	Virion morphogenesis protein OPG132	A6 is essential for virion membrane biogenesis and localization of virion membrane proteins to sites of virion assembly.

Pocket ID: P_1 PubChem CID: 3410901 Binding Affinity (kcal/mol): -8.601 ADMET Flag: Moderate	
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Table 14: Detailed intermolecular interaction profile of ligand 3410901 docked with protein AAL40575.1, including interaction distances (Å), bond categories, and associated van der Waals contacts within the predicted binding pocket.

Name	Distance (Å)	Category	Van der Waal's Interactions (VdW)
A:SER369:CB - 3410901:O	3.69	Hydrogen Bond	IleA:183
A:TYR204 - 3410901	5.01	Hydrophobic	IleA:313
A:LYS321 - 3410901	3.95	Hydrophobic	SerA:325
A:PHE186 - 3410901	4.47	Hydrophobic	PheA:364
A:SER369:CB - 3410901	3.84	Hydrophobic	TyrA:368
A:PHE372 - 3410901	5.01	Hydrophobic	
3410901 - A:LYS321	5.33	Hydrophobic	
3410901 - A:ILE324	5.18	Hydrophobic	

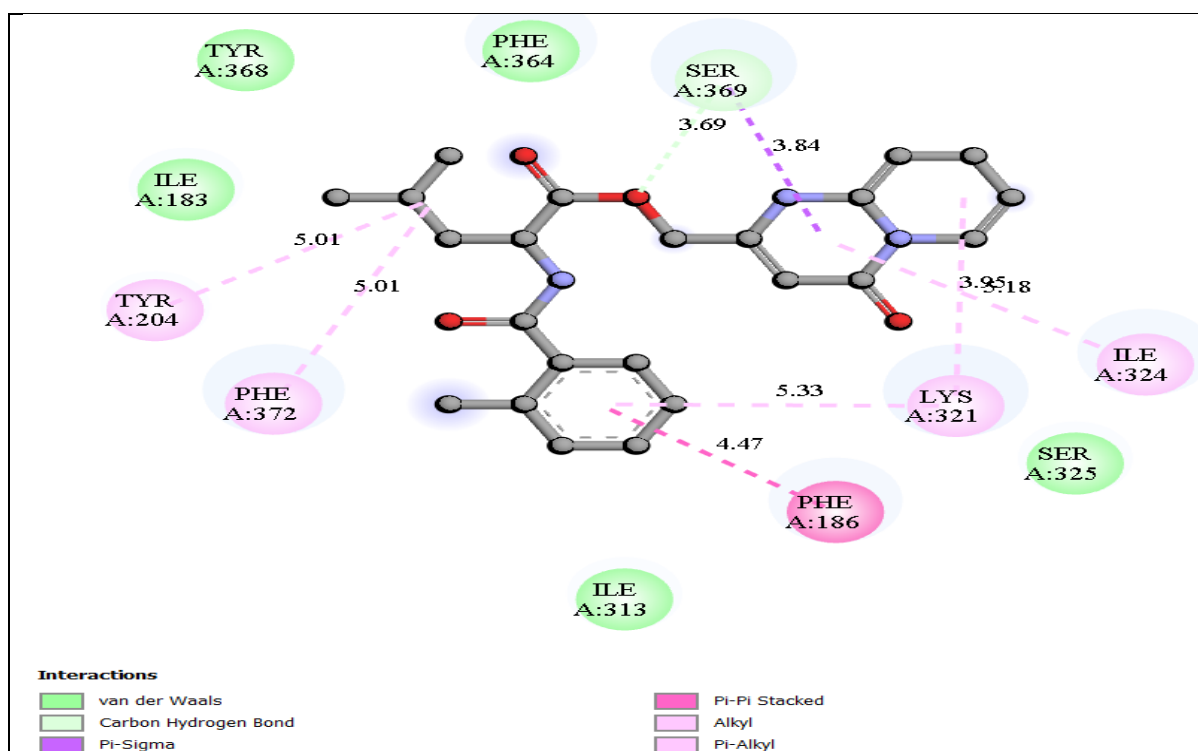
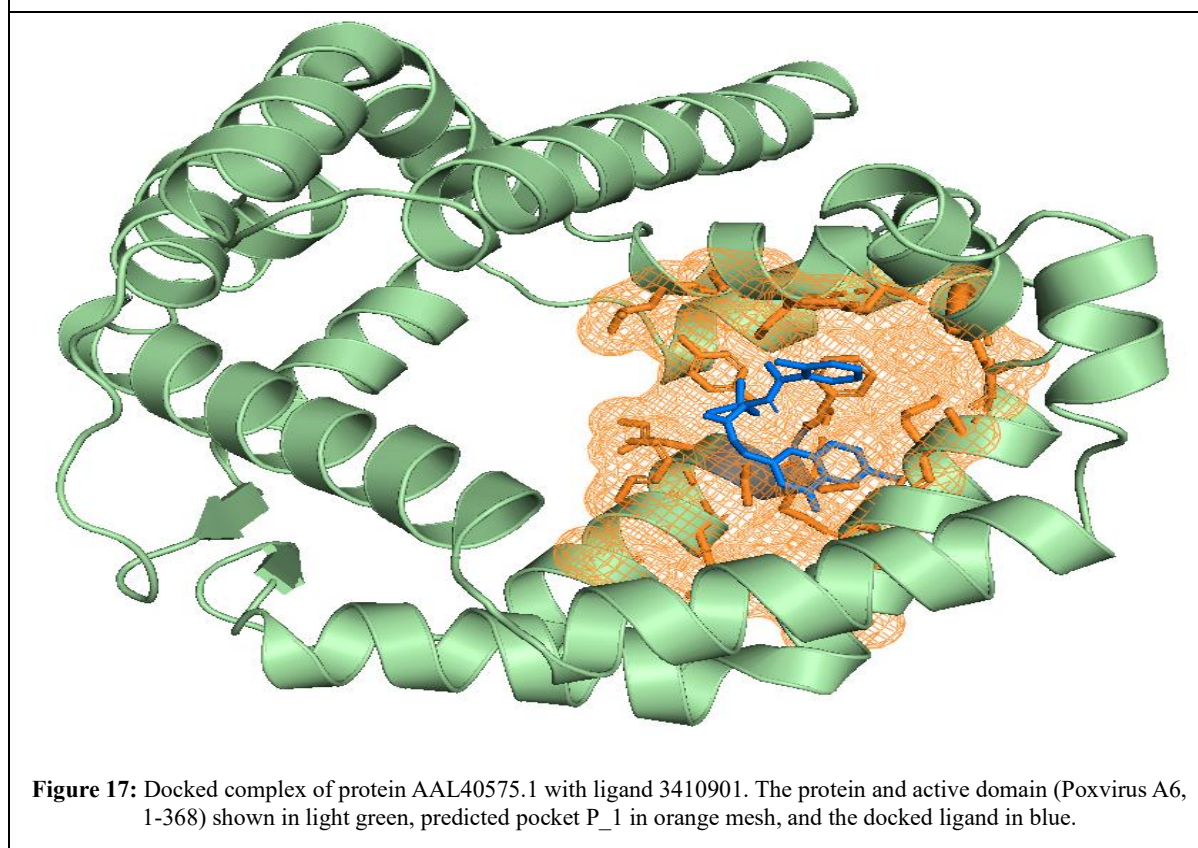


Figure 16: 2D protein–ligand interaction map of PubChem CID 3410901 within the predicted binding pocket of the Mpox protein AAL40575.1 (Virion morphogenesis protein OPG132), illustrating the key interacting residues.

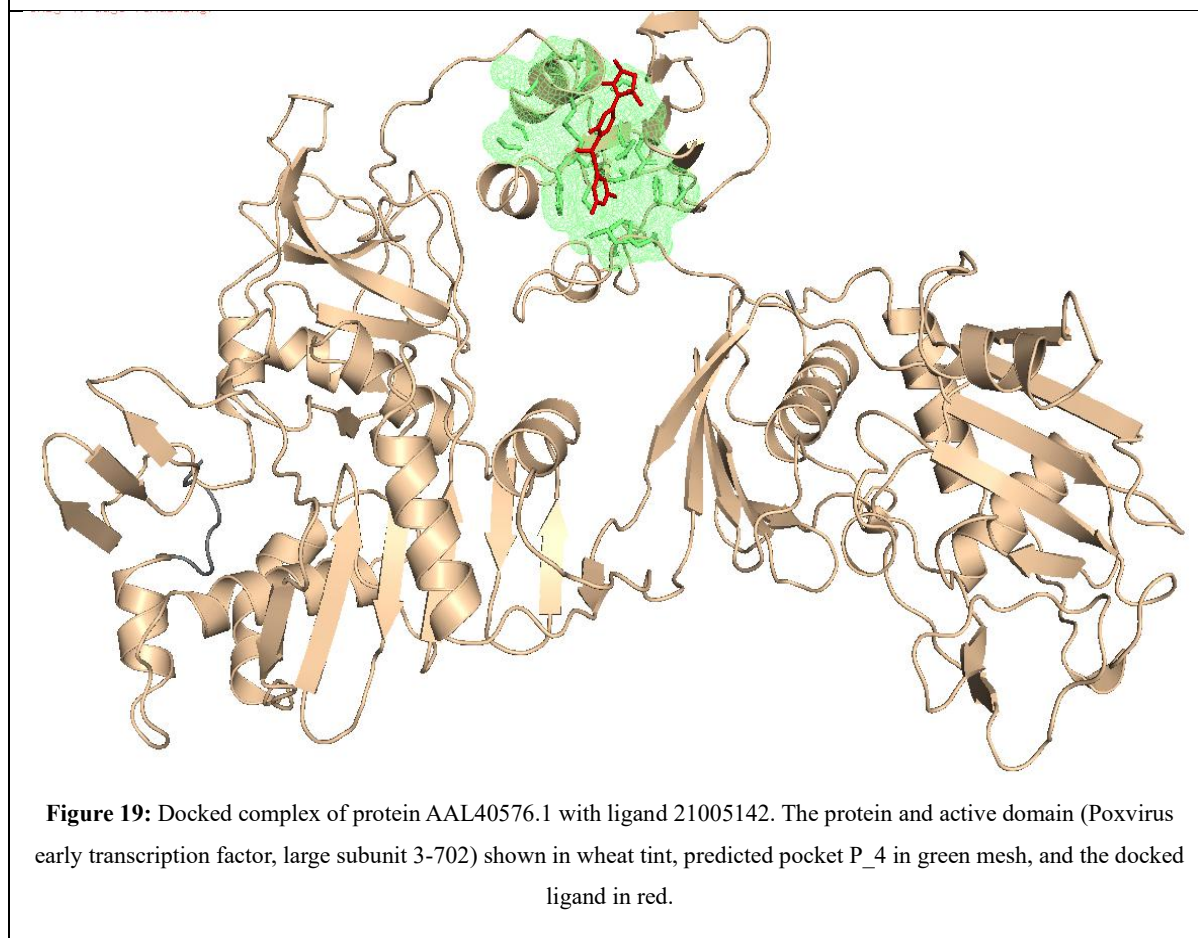
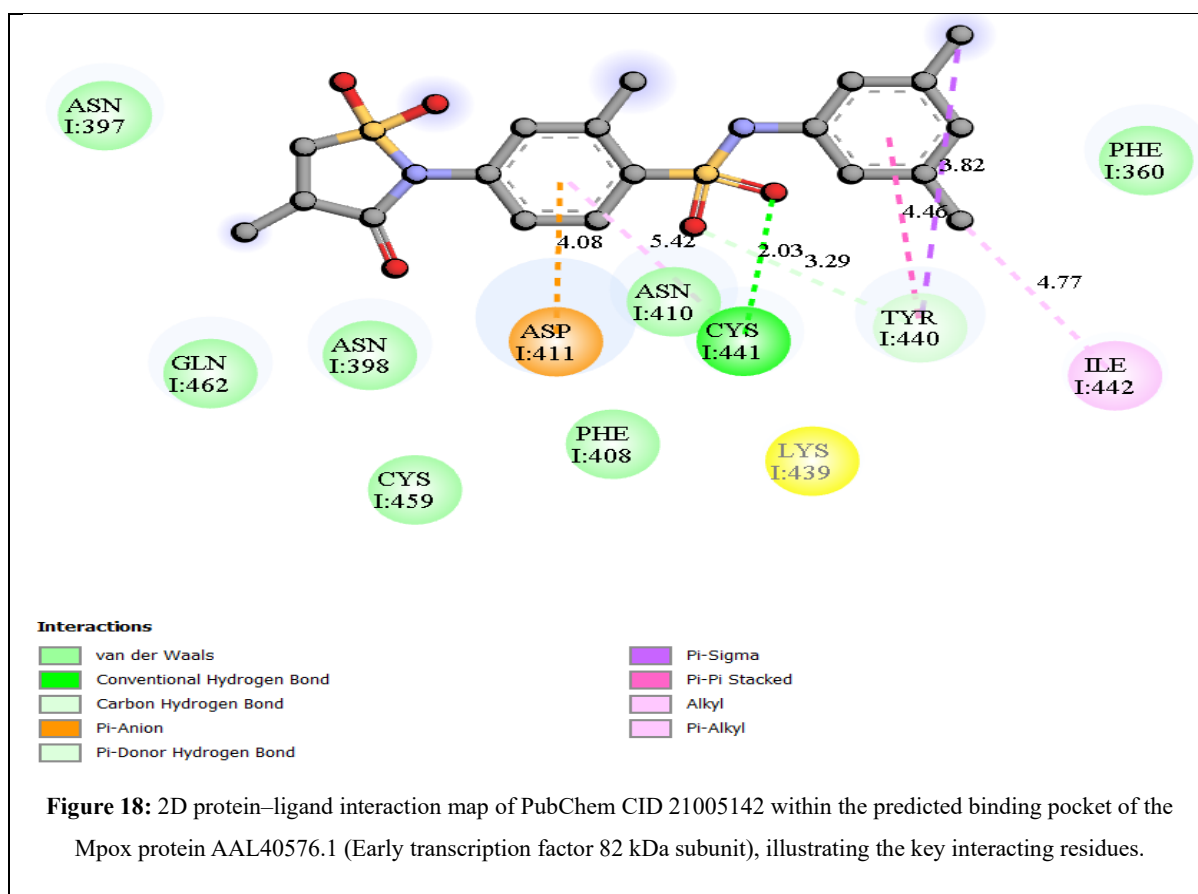


	Protein NCBI Accession ID	Protein Name	Protein Function
4)	AAL40576.1	Early transcription factor 82 kDa subunit	A viral transcription factor subunit. Initiates early gene transcription.

Pocket ID: P_4 PubChem CID: 21005142 Binding Affinity (kcal/mol): -8.934 ADMET Flag: Moderate	
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Table 15: Detailed intermolecular interaction profile of ligand 21005142 docked with protein AAL40576.1, including interaction distances (Å), bond categories, and associated van der Waals contacts within the predicted binding pocket.

Name	Distance (Å)	Category	Van der Waal's Interactions (VdW)
21005142:C - I:ILE442	4.77	Hydrophobic	PheI:360
I:CYS441:HN - 21005142:O	2.03	Hydrogen Bond	AsnI:397
I:ASP411:OD1 - 21005142	4.08	Electrostatic	AsnI:398
21005142:C - I:TYR440	3.82	Hydrophobic	PheI:408
I:TYR440 - 21005142	4.46	Hydrophobic	AsnI:410
21005142 - I:CYS441	5.42	Hydrophobic	TyrI:440
I:ASP411:HN - 21005142	3.11	Hydrogen Bond	CysI:459
I:TYR440:CA - 21005142:O	3.29	Hydrogen Bond	GlnI:462



CHAPTER-5

CONCLUSION

This study employed an integrated comparative genomics, subtractive proteomics, structural bioinformatics, and computer-aided drug discovery pipeline to identify conserved therapeutic targets and potential antiviral compounds against Mpox. A systematically designed multi-stage filtering strategy enabled the progressive refinement of proteins, binding pockets, and ligand candidates, thereby increasing the biological relevance and computational reliability of the final prioritized complexes.

Initially, comparative genomic analysis of 28 Mpox virus genomes representing multiple clades identified 129 conserved single-copy orthogroups, reflecting a highly stable core proteome across genetically diverse viral strains. Functional annotation and literature-based prioritization subsequently reduced this set to 27 biologically essential proteins involved in critical viral processes such as replication, transcription, virion assembly, and morphogenesis. Further subtractive proteomics analysis excluded proteins exhibiting significant similarity to the human proteome, resulting in a refined set of 22 non-host homologous proteins with improved therapeutic specificity and reduced likelihood of host off-target interactions.

Structural modelling and quality assessment further narrowed the candidate pool. Proteins lacking reliable structural templates or acceptable model quality were removed to ensure downstream structural analyses were based on stable and biologically meaningful protein conformations. Domain-based filtering and functional annotation subsequently refined the dataset to 14 proteins containing clearly defined functional regions suitable for pocket mapping and docking studies. Binding pocket analysis identified multiple cavities across these proteins; however, only pockets demonstrating favourable druggability, appropriate cavity volume, and strong overlap with annotated functional domains were retained. This refinement reduced the dataset to 35 biologically relevant binding pockets distributed across 13 proteins. A pocket-guided ligand selection strategy was then implemented using compounds retrieved from the ZINC12 and PubChem databases. Five ligands were selected for each retained pocket, resulting in an initial screening library of 175 compounds. Molecular docking analysis identified 36 ligands with strong binding affinities (≤ -8.0 kcal/mol) across 5 proteins and 13 binding pockets. Subsequent ADMET filtering further eliminated compounds associated with unfavourable pharmacokinetic behaviour, excessive predicted toxicity, or severe metabolic liabilities, ultimately reducing the candidate pool to 11 ligands associated with 8 binding

pockets. Redocking validation under refined docking conditions confirmed the stability and reproducibility of most prioritized complexes, while two ligands exhibiting reduced affinity under stricter docking parameters were excluded.

Finally, DFT-based Frontier Molecular Orbital analysis enabled quantum chemical characterization of the shortlisted compounds. Ligands exhibiting favourable HOMO–LUMO energy gaps, balanced hardness–softness profiles, and acceptable electronic stability were prioritized, whereas compounds showing comparatively rigid or less reactive electronic behaviour were removed. This final refinement step resulted in the identification of five lead compounds (PubChem IDs: 3131912, 2368032, 46360552, 3410901, and 21005142) interacting with functionally important regions of Poly(A) polymerase catalytic subunit, Core protein OPG073, Virion morphogenesis protein OPG132, and Early transcription factor 82 kDa subunit.

Overall, the progressive reduction from 129 conserved proteins to 4 target proteins along with 5 prioritized ligand candidates demonstrates the effectiveness of the integrated computational workflow in eliminating structurally unreliable, biologically irrelevant, weakly interacting, or pharmacokinetically unsuitable candidates at each stage of analysis. The identified ligand–protein complexes exhibited favourable binding stability, acceptable pharmacokinetic properties, and promising electronic characteristics, supporting their potential as candidate inhibitors against Mpox virus targets. Although experimental validation remains necessary, the present study establishes a robust computational framework for broad-spectrum antiviral target discovery and rational therapeutic development against Mpox virus.

SUPPLEMENTARY MATERIALS:

The supplementary materials associated with this study are provided as separate Excel files accompanying the thesis submission. These datasets contain detailed computational outputs, large-scale analysis results, and extended tabular information generated during the bioinformatics and computer-aided drug discovery workflow.

- **Supplementary Table S1:** Complete functional annotation dataset of 129 conserved Mpox viral proteins, including NCBI protein identifiers.
- **Supplementary Table S2:** Refined list of essential and non-homologous Mpox viral proteins, including NCBI IDs, UniProt IDs, functional annotations, FASTA sequences, and supporting literature references DOI.
- **Supplementary Table S3:** Detailed binding pocket characterization data, including pocket volume, surface area, druggability score, depth, hydrophobicity, amino acid composition, hydrogen bonding properties, atom distribution parameters, druggability class.
- **Supplementary Table S4:** Curated ligand dataset retrieved from ZINC12 and PubChem, including PubChem Compound IDs (CID) and SMILES representations of shortlisted ligands selected through pocket-guided filtering.
- **Supplementary Table S5:** Complete molecular docking results of screened ligands, including protein targets, pocket IDs, docking grid dimensions, ligand PubChem IDs, and binding affinity values.
- **Supplementary Table S6:** ADMET profiling dataset containing physicochemical properties, pharmacokinetic predictions, toxicity assessment, and ADMET risk categorization of ligands satisfying the docking threshold and SMILES.
- **Supplementary Table S7:** Final prioritized ligand candidates retained after integrated docking and ADMET filtering, including shortlisted protein–ligand complexes, binding affinity, physiochemical, toxicity properties, ADMET description and ADMET flag selected for redocking and interaction validation studies.

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APPENDIX:

Python Script for DFT-Based FMO Analysis:

The following Python script was used to perform density functional theory (DFT)-based analysis of selected compounds, including HOMO–LUMO energy calculations, molecular electrostatic potential (MEP) mapping, dipole, and global reactivity descriptor evaluation. The workflow integrates RDKit for molecular structure generation and optimization, PySCF for quantum chemical computations, and py3Dmol for three-dimensional visualization.

```
!pip install rdkit pyscf py3Dmol
import numpy as np
import matplotlib.pyplot as plt
import py3Dmol
import time
import os
from rdkit import Chem
from rdkit.Chem import AllChem
from rdkit.Chem.rdmolops import GetFormalCharge
from pyscf import gto
from pyscf.tools import cubegen

# Attempt to import GPU support; fall back to CPU if not available
try:
    from gpu4pyscf import dft as gpu_dft
    dft = gpu_dft
    print("Using GPU for DFT via gpu4pyscf")
except ImportError:
    from pyscf import dft
    print("gpu4pyscf not installed; using CPU for DFT. Install with: !pip install
gpu4pyscf")

def compute_mep(pyscf_mol, mf, label):
    try:
        os.makedirs('/kaggle/working/orbitals', exist_ok=True)
        mep_file = f'/kaggle/working/orbitals/{label}_mep.cube'
        cubegen.density(pyscf_mol, mep_file, mf.make_rdm1())
        print(f"MEP cube file saved: {mep_file}")
        return mep_file
    except Exception as e:
        print(f"Error generating MEP for {label}: {e}")
        return None

def compute_dipole(pyscf_mol, mf, label):
    try:
        dipole = mf.dip_moment(unit='Debye')
        print(f"{label} Dipole Moment (Debye): {dipole}")
        return dipole
    except Exception as e:
        print(f"Error computing dipole for {label}: {e}")
        return None

def compute_reactivity_descriptors(homo_ev, lumo_ev, label):
    try:
        hardness = (lumo_ev - homo_ev) / 2
        softness = 1 / hardness if hardness != 0 else float('inf')
        electronegativity = -(homo_ev + lumo_ev) / 2
        electrophilicity = (electronegativity ** 2) / (2 * hardness) if hardness != 0 else
float('inf')
        print(f"{label} Reactivity Descriptors:")
        print(f"Chemical Hardness ( $\eta$ ): {hardness:.4f} eV")
```

```

        print(f"Chemical Softness (1/η): {softness:.4f} eV-1")
        print(f"Electronegativity (χ): {electronegativity:.4f} eV")
        print(f"Electrophilicity (ω): {electrophilicity:.4f} eV")
        return hardness, softness, electronegativity, electrophilicity
    except Exception as e:
        print(f"Error computing reactivity descriptors for {label}: {e}")
        return None, None, None, None

def save_summary(label, homo_ev, lumo_ev, gap_ev, dipole, hardness, softness,
electronegativity, electrophilicity):
    try:
        os.makedirs('/kaggle/working/orbitals', exist_ok=True)
        with open(f'/kaggle/working/orbitals/{label}_summary.txt', 'w') as f:
            f.write(f"{label} Computational Summary\n")
            f.write(f"HOMO: {homo_ev:.4f} eV\n")
            f.write(f"LUMO: {lumo_ev:.4f} eV\n")
            f.write(f"Gap: {gap_ev:.4f} eV\n")
            if dipole is not None:
                f.write(f"Dipole Moment: {dipole}\n")
            if hardness is not None:
                f.write(f"Chemical Hardness: {hardness:.4f} eV\n")
                f.write(f"Chemical Softness: {softness:.4f} eV-1\n")
                f.write(f"Electronegativity: {electronegativity:.4f} eV\n")
                f.write(f"Electrophilicity: {electrophilicity:.4f} eV\n")
            print(f"Summary saved to /kaggle/working/orbitals/{label}_summary.txt")
    except Exception as e:
        print(f"Error saving summary for {label}: {e}")

def compute_homo_lumo_and_save(smiles, label, basis='sto-3g'):
    start_time = time.time()

    # Parse SMILES and compute charge
    mol = Chem.MolFromSmiles(smiles)
    if mol is None:
        print(f"Invalid SMILES for {label}")
        return False
    charge = GetFormalCharge(mol)
    num_atoms = mol.GetNumAtoms()
    print(f"Processing {label}: {num_atoms} atoms, charge {charge}")

    # Generate 3D conformation with robust embedding
    mol = Chem.AddHs(mol)
    embed_params = AllChem.ETKDGv3()
    embed_params.maxIterations = 1000
    embed_params.randomSeed = 42
    embed_params.useRandomCoords = True
    embed_status = AllChem.EmbedMolecule(mol, embed_params)

    if embed_status != 0:
        print(f"Initial embedding failed for {label}. Trying UFF optimization...")
        try:
            AllChem.UFFOptimizeMolecule(mol, maxIters=1000)
            if mol.GetConformer().Is3D():
                print("UFF optimization successful.")
            else:
                print(f"UFF optimization failed to produce 3D structure for {label}.")
                return False
        except Exception as e:
            print(f"UFF optimization failed for {label}: {e}")
            print("Consider using external tools like OpenBabel or CREST.")
            return False

    # Optimize with MMFF
    try:
        AllChem.MMFFOptimizeMolecule(mol, maxIters=1000)
        print("MMFF optimization successful.")

```



```

except Exception as e:
    print(f"MMFF optimization failed for {label}: {e}, proceeding with current
geometry.")

# Extract XYZ coordinates for DFT
conf = mol.GetConformer()
xyz = ''
for atom in mol.GetAtoms():
    pos = conf.GetAtomPosition(atom.GetIdx())
    xyz += f"{atom.GetSymbol()} {pos.x:.4f} {pos.y:.4f} {pos.z:.4f}\n"

# Build PySCF molecule
pyscf_mol = gto.M(atom=xyz, basis=basis, charge=charge, spin=0)

# Run DFT (B3LYP)
mf = dft.RKS(pyscf_mol)
mf.xc = 'b3lyp'
mf.conv_tol = 1e-8
mf.max_cycle = 200
try:
    energy = mf.kernel()
    if not mf.converged:
        print(f"DFT calculation did not converge for {label}. Trying PBE
functional...")
        mf.xc = 'pbe'
        energy = mf.kernel()
        if not mf.converged:
            print(f"DFT calculation failed for {label} with PBE.")
            return False
except Exception as e:
    print(f"DFT calculation failed for {label}: {e}")
    return False

# Compute dipole moment
dipole = compute_dipole(pyscf_mol, mf, label)

# Generate MEP
mep_file = compute_mep(pyscf_mol, mf, label)

# Get HOMO and LUMO indices
mo_energy = mf.mo_energy
mo_occ = mf.mo_occ
homo_idx = None
lumo_idx = None
for i in range(len(mo_occ)):
    if mo_occ[i] > 0:
        homo_idx = i
    elif mo_occ[i] == 0 and lumo_idx is None:
        lumo_idx = i
if homo_idx is None or lumo_idx is None:
    print(f"Failed to identify HOMO/LUMO for {label}. Orbital occupancy issue.")
    return False

homo_ev = mo_energy[homo_idx] * 27.211386
lumo_ev = mo_energy[lumo_idx] * 27.211386
gap_ev = lumo_ev - homo_ev

# Check for valid HOMO/LUMO energies
if abs(gap_ev) < 1e-6:
    print(f"Warning: HOMO and LUMO energies are identical for {label}. Possible DFT
convergence issue.")
    return False

# Compute reactivity descriptors
hardness, softness, electronegativity, electrophilicity =
compute_reactivity_descriptors(homo_ev, lumo_ev, label)

```

```

# Save cube files for HOMO and LUMO
os.makedirs('/kaggle/working/orbitals', exist_ok=True)
homo_cube_file = f'/kaggle/working/orbitals/{label}_homo.cube'
lumo_cube_file = f'/kaggle/working/orbitals/{label}_lumo.cube'
try:
    cubegen.orbital(pyscf_mol, homo_cube_file, mf.mo_coeff[:, homo_idx])
    cubegen.orbital(pyscf_mol, lumo_cube_file, mf.mo_coeff[:, lumo_idx])
    if not (os.path.exists(homo_cube_file) and os.path.exists(lumo_cube_file)):
        print(f"Failed to create cube files for {label}.")
        return False
except Exception as e:
    print(f"Error generating cube files for {label}: {e}")
    return False

# Save summary
save_summary(label, homo_ev, lumo_ev, gap_ev, dipole, hardness, softness,
electronegativity, electrophilicity)

# Print results
end_time = time.time()
print(f"{label} results (basis: {basis}):")
print(f"HOMO: {homo_ev:.4f} eV")
print(f"LUMO: {lumo_ev:.4f} eV")
print(f"Gap: {gap_ev:.4f} eV")
print(f"Time: {end_time - start_time:.2f} seconds")
print(f"Cube files saved: {homo_cube_file}, {lumo_cube_file}\n")

# Visualize molecule with Py3DMol (structure only)
mol_block = Chem.MolToMolBlock(mol)
viewer = py3Dmol.view(width=400, height=400)
viewer.addModel(mol_block, 'mol')
viewer.setStyle({'stick': {}})
viewer.zoomTo()
viewer.show()

return True

def read_cube_file(filename):
    try:
        with open(filename, 'r') as f:
            lines = f.readlines()
        if len(lines) < 6:
            raise ValueError(f"Cube file {filename} is too short or corrupted.")
        natoms = int(lines[2].split()[0])
        origin = np.array([float(x) for x in lines[2].split()[1:4]])
        nx, ny, nz = [int(lines[i].split()[0]) for i in range(3, 6)]
        voxel = np.array([[float(lines[i].split()[j]) for j in range(1, 4)] for i in
range(3, 6)])
        if nx <= 0 or ny <= 0 or nz <= 0:
            raise ValueError(f"Invalid dimensions in cube file {filename}: nx={nx},
ny={ny}, nz={nz}")
        data = []
        for line in lines[6 + natoms:]:
            data.extend([float(x) for x in line.split()])
        data = np.array(data).reshape((nx, ny, nz))
        if np.all(data == 0):
            print(f"Warning: Volumetric data in {filename} is all zeros. Orbital data may
be invalid.")
        if np.any(np.isnan(data)) or np.any(np.isinf(data)):
            raise ValueError(f"Invalid volumetric data in {filename}: contains NaN or Inf
values.")
        x = np.linspace(0, nx-1, nx) * voxel[0][0] + origin[0]
        y = np.linspace(0, ny-1, ny) * voxel[1][1] + origin[1]
        z = np.linspace(0, nz-1, nz) * voxel[2][2] + origin[2]
        X, Y, Z = np.meshgrid(x, y, z, indexing='ij')

```

```

        return X, Y, Z, data
    except Exception as e:
        print(f"Error reading cube file {filename}: {e}")
        return None, None, None, None

def plot_2d_slice(X, Y, data, label, plane='xy', z_slice=0):
    if X is None or Y is None or data is None:
        print(f"Skipping 2D plot for {label} due to invalid cube data.")
        return
    plt.figure(figsize=(8, 6))
    try:
        if plane == 'xy':
            z_idx = np.argmin(np.abs(Z[0, 0, :] - z_slice))
            slice_data = data[:, :, z_idx]
            plt.contourf(X[:, :, z_idx], Y[:, :, z_idx], slice_data, levels=20,
cmap='RdBu')
            plt.colorbar(label='Orbital amplitude')
            plt.xlabel('X (Angstrom)')
            plt.ylabel('Y (Angstrom)')
            plt.title(f'{label} 2D Slice ({plane} plane, z={z_slice:.2f} Å)')
            os.makedirs('/kaggle/working/orbitals', exist_ok=True)
            plt.savefig(f'/kaggle/working/orbitals/{label}_2d_slice.png')
            plt.show()
        except Exception as e:
            print(f"Error plotting 2D slice for {label}: {e}")

def visualize_3d_orbital(cube_file, label, smiles, isovalues=[0.01, 0.02, 0.03]):
    try:
        with open(cube_file, 'r') as f:
            cube_data = f.read()
        viewer = py3Dmol.view(width=400, height=400)
        for isoval in isovalues:
            viewer.addVolumetricData(cube_data, 'cube', {'isoval': isoval, 'color': 'blue',
'opacity': 0.8})
            viewer.addVolumetricData(cube_data, 'cube', {'isoval': -isoval, 'color': 'red',
'opacity': 0.8})
        mol = Chem.MolFromSmiles(smiles)
        if mol is None:
            print(f"Failed to parse SMILES for {label}")
            return
        mol = Chem.AddHs(mol)
        embed_params = AllChem.ETKDGv3()
        embed_params.maxIterations = 2000
        embed_params.randomSeed = 42
        for attempt in range(5):
            embed_status = AllChem.EmbedMolecule(mol, embed_params)
            if embed_status == 0 and mol.GetConformer().Is3D():
                print(f"Embedding successful for {label} on attempt {attempt + 1}")
                break
            embed_params.randomSeed += 1
        else:
            print(f"Embedding failed for {label} after 5 attempts. Using UFF
optimization...")
            try:
                AllChem.UFFOptimizeMolecule(mol, maxIters=2000)
                if not mol.GetConformer().Is3D():
                    print(f"UFF optimization failed for {label}. Skipping 3D structure.")
                    viewer.show()
                    return
            except Exception as e:
                print(f"UFF optimization failed for {label}: {e}. Skipping 3D structure.")
                viewer.show()
                return
            try:
                AllChem.MMFFOptimizeMolecule(mol, maxIters=2000)
                print(f"MMFF optimization successful for {label}")
    
```

```

        except Exception as e:
            print(f"MMFF optimization failed for {label}: {e}. Using current geometry.")
            mol_block = Chem.MolToMolBlock(mol)
            viewer.addModel(mol_block, 'mol')
            viewer.setStyle({'stick': {}})
            viewer.zoomTo()
            viewer.show()
            print(f"3D visualization for {label} displayed with isovalues {isovalues}.")
    except Exception as e:
        print(f"Error in 3D visualization for {label}: {e}")

# Main execution
smiles = r'COC1=CC(=CC(=C1OC)OC)C2=NC3=C(C=NN3C(=C2)C(F)(F)F)C(=O)NCC4=CC=CO4'  #SMILES of
compound
label = '3131912'  #Output Folder Name

success = compute_homo_lumo_and_save(smiles, label, basis='sto-3g')

if success:
    homo_file = f'/kaggle/working/orbitals/{label}_homo.cube'
    lumo_file = f'/kaggle/working/orbitals/{label}_lumo.cube'
    mep_file = f'/kaggle/working/orbitals/{label}_mep.cube'

    # Read and plot HOMO
    X, Y, Z, homo_data = read_cube_file(homo_file)
    if homo_data is not None:
        plot_2d_slice(X, Y, homo_data, f'{label}_homo', plane='xy', z_slice=0)
        visualize_3d_orbital(homo_file, f'{label}_homo', smiles, isovalues=[0.01, 0.02,
0.03])

    # Read and plot LUMO
    X, Y, Z, lumo_data = read_cube_file(lumo_file)
    if lumo_data is not None:
        plot_2d_slice(X, Y, lumo_data, f'{label}_lumo', plane='xy', z_slice=0)
        visualize_3d_orbital(lumo_file, f'{label}_lumo', smiles, isovalues=[0.01, 0.02,
0.03])

    # Visualize MEP
    if os.path.exists(mep_file):
        visualize_3d_orbital(mep_file, f'{label}_mep', smiles, isovalues=[0.001, 0.002])

    # Debug: List files in output directory
    output_dir = '/kaggle/working/orbitals'
    print("Final files in output 'orbitals' directory:", os.listdir(output_dir) if
os.path.exists(output_dir) else "Directory not created")
else:
    print("Computation failed, skipping visualization.")

```